



ENDUROSAT

# **Electrical Power System EPS I & EPS I Plus**

## **User Manual**

Version: 7.12

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## Document Scope

This User Manual contains comprehensive information regarding the Electric Power System (EPS) I and EPS I plus, hardware 1.7 and firmware 1678 version. It highlights the features of the module, includes a guide to its proper handling and usage, and provides a detailed system description on a mechanical, hardware and software level. The instructions included herein should allow you to successfully operate the EPS I and interface it with the remaining modules of your satellite. If you require further assistance, please contact EnduroSat's Customer Support.

|                 | Version number |
|-----------------|----------------|
| <b>Hardware</b> | 1.7            |
| <b>Firmware</b> | 16.78          |

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## Contents

|                                               |           |
|-----------------------------------------------|-----------|
| <b>List of Tables</b>                         | <b>6</b>  |
| <b>List of Figures</b>                        | <b>7</b>  |
| <b>List of Acronyms</b>                       | <b>8</b>  |
| <b>Revision History</b>                       | <b>9</b>  |
| <b>1 Safety Precautions and Certification</b> | <b>10</b> |
| 1.1 Handling Instructions . . . . .           | 10        |
| 1.2 Warnings . . . . .                        | 10        |
| 1.3 Quality Standards . . . . .               | 12        |
| 1.4 Qualification Tests . . . . .             | 12        |
| <b>2 General Overview</b>                     | <b>13</b> |
| 2.1 Key Features . . . . .                    | 13        |
| <b>3 System Description</b>                   | <b>15</b> |
| 3.1 Mechanical Description . . . . .          | 21        |
| 3.2 Connectivity . . . . .                    | 23        |
| 3.2.1 Connectors & Pinouts . . . . .          | 23        |
| <b>4 System Functionality</b>                 | <b>32</b> |
| 4.1 Electrical Characteristics . . . . .      | 32        |
| 4.2 EPS Module Monitoring . . . . .           | 35        |
| 4.3 EPS Module Control . . . . .              | 36        |



|          |                                            |           |
|----------|--------------------------------------------|-----------|
| 4.4      | Firmware Update . . . . .                  | 39        |
| <b>5</b> | <b>System Software</b>                     | <b>40</b> |
| 5.1      | ESPS I . . . . .                           | 40        |
| 5.2      | Module Interfaces . . . . .                | 41        |
| <b>6</b> | <b>Software Commands</b>                   | <b>42</b> |
| 6.1      | EPS I Interface Methods Overview . . . . . | 42        |
| 6.2      | Data Types . . . . .                       | 44        |
| 6.3      | Interface Methods . . . . .                | 54        |
| 6.3.1    | getSoftwareVersion (id: 0x00) . . . . .    | 54        |
| 6.3.2    | getBatteryInfo (id: 0x01) . . . . .        | 55        |
| 6.3.3    | getBusesInfo (id: 0x02) . . . . .          | 55        |
| 6.3.4    | getSolarPanelsInfo (id: 0x03) . . . . .    | 56        |
| 6.3.5    | getStatisticInfo (id: 0x04) . . . . .      | 56        |
| 6.3.6    | getTemperaturesInfo (id: 0x05) . . . . .   | 57        |
| 6.3.7    | getConfigurationInfo (id: 0x06) . . . . .  | 57        |
| 6.3.8    | queryControls (id: 0x07) . . . . .         | 58        |
| 6.3.9    | queryRBatRaw (id: 0x08) . . . . .          | 59        |
| 6.3.10   | queryDefaults (id: 0x09) . . . . .         | 60        |
| 6.3.11   | queryLaunchState (id: 0x0A) . . . . .      | 62        |
| 6.3.12   | queryWatchdog (id: 0x0B) . . . . .         | 63        |
| 6.3.13   | resetWatchdog (id: 0x0C) . . . . .         | 64        |
| 6.3.14   | resetPowerSupply (id: 0x0D) . . . . .      | 65        |
| 6.3.15   | SetSerialNumber (id: 0x0E) . . . . .       | 66        |



|        |                                                   |    |
|--------|---------------------------------------------------|----|
| 6.3.16 | GetSerialNumber (id: 0x0F) . . . . .              | 67 |
| 6.3.17 | HIBModeRead (id: 0x10) . . . . .                  | 67 |
| 6.3.18 | HIBModeWrite (id: 0x11) . . . . .                 | 68 |
| 6.3.19 | HeatersSettingsRead (id: 0x12) . . . . .          | 70 |
| 6.3.20 | HeatersSettingsWrite (id: 0x13) . . . . .         | 71 |
| 6.3.21 | ToggleOutPut (id: 0x14) . . . . .                 | 72 |
| 6.3.22 | NotificationsSettings (id: 0x15) . . . . .        | 73 |
| 6.3.23 | GetTelemetryAltogether (id: 0x16) . . . . .       | 74 |
| 6.3.24 | bload_getVersion (id: 0x10000000) . . . . .       | 75 |
| 6.3.25 | bload_eraseFlash (id: 0x10000001) . . . . .       | 75 |
| 6.3.26 | bload_writeFlash (id: 0x10000002) . . . . .       | 76 |
| 6.3.27 | bload_execute (id: 0x10000003) . . . . .          | 77 |
| 6.4    | EPS I Notify Interface Methods Overview . . . . . | 78 |
| 6.5    | Data Types . . . . .                              | 78 |
| 6.6    | Interface Methods . . . . .                       | 79 |
| 6.6.1  | EventsNotification (id: 0x01) . . . . .           | 79 |



## List of Tables

|    |                                                              |    |
|----|--------------------------------------------------------------|----|
| 2  | Comparison of the EPS I and the EPS I Plus modules . . . . . | 17 |
| 3  | Pinouts of H1 connector . . . . .                            | 25 |
| 4  | Pinouts of H2 connector . . . . .                            | 26 |
| 5  | Pinouts of PUSB Connector. . . . .                           | 27 |
| 6  | PUSB Connector Specifications . . . . .                      | 28 |
| 7  | EPS I Solar Panel Connectors and Associated Axis. . . . .    | 29 |
| 8  | Pinouts of each EPS I Solar Panel Connector. . . . .         | 29 |
| 9  | EPS I Solar Panel Connectors Specifications . . . . .        | 29 |
| 10 | RBF Connector Specifications . . . . .                       | 30 |
| 11 | Kill Switches Specifications . . . . .                       | 31 |
| 12 | Electrical Characteristics . . . . .                         | 32 |
| 13 | Module Interfaces . . . . .                                  | 41 |



## List of Figures

|    |                                                                                                          |    |
|----|----------------------------------------------------------------------------------------------------------|----|
| 1  | EPS I module . . . . .                                                                                   | 15 |
| 2  | Block diagram of the EPS I Module . . . . .                                                              | 16 |
| 3  | Algorithm for protection of the batteries from short circuit, deep discharge and<br>overheating. . . . . | 19 |
| 4  | Algorithm for optimization of heater power consumption. . . . .                                          | 20 |
| 5  | EPS I (includes 1 battery pack) . . . . .                                                                | 21 |
| 6  | EPS I Plus (includes 2 battery packs) . . . . .                                                          | 22 |
| 7  | EPS I module top view . . . . .                                                                          | 23 |
| 8  | EPS I module side view . . . . .                                                                         | 23 |
| 9  | EPS PC104 pinout diagram . . . . .                                                                       | 24 |
| 10 | USB-SCIC connector. . . . .                                                                              | 27 |
| 11 | EPS Solar Panel connector. . . . .                                                                       | 28 |
| 12 | RBF Connector . . . . .                                                                                  | 30 |
| 13 | RBF and Kill Switches pins . . . . .                                                                     | 30 |
| 14 | Kill Switches Connectors . . . . .                                                                       | 31 |
| 15 | Algorithm for the device power-on sequence . . . . .                                                     | 38 |





## List of Acronyms

|                       |                                             |
|-----------------------|---------------------------------------------|
| <b>BCR</b>            | Battery Charger Regulator                   |
| <b>CC-CV</b>          | Constant Current - Constant Voltage         |
| <b>EOC</b>            | End of Charge                               |
| <b>EPS</b>            | Electric Power System                       |
| <b>ESD</b>            | Electrostatic Discharge                     |
| <b>ESPS I</b>         | EnduroSat Protocol Stack I                  |
| <b>GND</b>            | Ground                                      |
| <b>GPIO</b>           | General-Purpose Input/Output                |
| <b>GPO</b>            | General Purpose Output                      |
| <b>I<sup>2</sup>C</b> | Inter-Integrated Circuit                    |
| <b>ISS</b>            | International Space Station                 |
| <b>Li-Po</b>          | Lithium Polymer                             |
| <b>LUP</b>            | Latch-up Protection                         |
| <b>MCU</b>            | Microcontroller Unit                        |
| <b>MPPT</b>           | Maximum Power Point Tracking                |
| <b>OBC</b>            | On-board Computer                           |
| <b>PCM</b>            | Protection Circuit Module                   |
| <b>PUSB</b>           | Powered Universal Serial Bus                |
| <b>RBF</b>            | Remove Before Flight                        |
| <b>RH</b>             | Relative Humidity                           |
| <b>RTC</b>            | Real Time Clock                             |
| <b>SCIC</b>           | Satellite Communication Interface Connector |
| <b>SP</b>             | Solar Panel                                 |
| <b>USB</b>            | Universal Serial Bus                        |
| <b>VRTC</b>           | Voltage Output for Real Time Clock          |



## Revision History

| Date                     | Revision | Author(s)                              | Page(s)        | Reason for Change                                                  |
|--------------------------|----------|----------------------------------------|----------------|--------------------------------------------------------------------|
| 20/03/2016               | 1.0      |                                        |                | Initial revision                                                   |
| 15/11/2018               | 2.0      | Evangelia Gkaravela                    | multiple pages | Updated version                                                    |
| 10/04/2021               | 3.0      | Evangelia Gkaravela                    | multiple pages | Document class change and various updates                          |
| 10/12/2021               | 4.0      | Evangelia Gkaravela                    | multiple pages | Document class change and various updates                          |
| 15/07/2022               | 5.0      | Evangelia Gkaravela                    | 42             | 6 Software Commands updated                                        |
| 30/08/2022               | 6.0      | Velislava Slavkova                     | 12             | 1.4 Qualification Tests updated                                    |
| 01/09/2022<br>01/02/2024 | 7.0-7.10 | Manfred Hamburgo<br>Velislava Slavkova | multiple pages | ESTTC depreciation, Software Commands updated and multiple updates |
| 26/03/2024               | 7.11     | Velislava Slavkova                     | 42             | 6 Software Commands updated                                        |
| 19/04/2024               | 7.12     | Velislava Slavkova                     | 42             | 6 Software Commands updated                                        |



# 1 Safety Precautions and Certification

## 1.1 Handling Instructions

Particular attention shall be paid to avoiding damage of the EPS I during storage and handling. Please observe the following instructions:

- Handle using gloves suitable for a clean room environment.
- The module should be handled in an environment which meets the requirements for a clean room class 100,000 (ISO 8), free of contaminants such as dust, oil, grease, fumes and smoke from any source.
- Store in such a manner as to avoid stress and prevent damage.
- To prevent deterioration, the EPS must be stored in a controlled environment (i.e. the environmental conditions must be maintained within the prescribed ranges):
  - Ideal storage temperature range: 15–27 °C.
  - Ideal storage humidity range: 30–60 % Relative Humidity (RH)

## 1.2 Warnings



Sensitive electronic device. Do not ship or store near strong electrostatic, electromagnetic, magnetic or radioactive fields.



This product uses semiconductors that can be damaged by electrostatic discharge. Observe precautions for handling.



Rechargeable lithium ion battery inside. May explode if exposed to fire.

- Do not incinerate or place near an open flame;
- Do not drop, crush, puncture, or disassemble;
- Do not short terminals;
- Do not expose to temperatures above 140 °F/ 60 °C
- Replace only with a battery specified by the manufacturer.



Do not throw away with household waste.



### 1.3 Quality Standards

The production process follows the quality standards:

- PC-A-610E, class 3 (Acceptability of Electronic Assemblies)
- IPC-A-600 (Acceptability of Printed Boards)
- J-STD-001 (Requirements for Soldered Electrical and Electronic Assemblies)
- ISO 14644 (Cleanrooms and Associated Controlled Environments)
- IEC 61340 (Electrostatics ESD: Protection of Electronic Devices from Electrostatic Phenomena)

### 1.4 Qualification Tests

A full campaign of tests at qualification level was performed on the EPS qualification engineering model. Qualification test levels and duration follow the ESA standard ECSS-E-ST-10-03C and GEVS: GSFC-STD-7000A. Environmental and Mechanical Tests performed:

- Thermal Cycling
- Thermal Vacuum
- Random Vibration
- Sine Vibration
- Shock Test
- Total Ionizing Dose



## 2 General Overview

EnduroSat's Electric Power System (EPS) I serves several critical roles within a nanosatellite – it (1) stores, transfers, and distributes the power, generated by the solar panels, throughout the system in order to keep it operational, and (2) provides a number of General Purpose Outputs (GPOs) for system control. It is characterized by very high efficiency and is designed in a way that makes it suitable for versatile use. The module has flight heritage and is fully compliant with both the CubeSat standard and the International Space Station (ISS) flight safety requirements. The EPS I incorporates firmware and hardware battery protection algorithms against short circuit, deep discharge and overheating. It connects to other modules via PC104 and to the solar panels via dedicated connectors.

The EPS I module has an additional variant – the EPS I Plus – which is functionally identical, but packs twice the battery power capacity. EnduroSat's EPS I and EPS I Plus modules are best suited for 1U, 1.5U and 2U CubeSat satellites.

### 2.1 Key Features

- Three solar panel input channels (one for each CubeSat axis: X, Y and Z)
- Input voltage (per solar panel channel): up to 5.5 V
- Input current (per solar panel channel): up to 1.8 A
- Six connectors for the solar panels
- Integrated blocking diode for each solar panel connector
- Battery Pack power capacity: 10.2 Wh (EPS I) or 20.4 Wh (EPS I Plus)
- Battery Pack voltage: 3.7 V nominal
- Stackable Battery Packs: up to 8 A
- Three battery charge modes : 230/460/690 mA
- Very low power consumption during normal operation: 20 mA @ 3.7 V battery
- Power buses: 3.3 V, 5 V, Battery Charger Regulator (BCR) (5 V max), Battery RAW, Real Time Clock (RTC) power BUS
- 3.3 V and 5 V Latch-up Protection (LUP) outputs
- Interfaces: RS-485, USB-C (Virtual COM Port)
- Protocol: ESPS I
- Two deployment and one Remove Before Flight (RBF) switches can be connected
- Six GPOs for on/off and reset of external modules
- USB for monitoring, control and battery charging



- Mass: 229.5 g (EPS I - includes 1 battery pack) or 329.7 g (EPS I Plus - includes 2 battery packs)



**Note:** Custom versions of the module with I<sup>2</sup>C and UART are offered upon customer demand.



### 3 System Description

Figures 1 and 2 show a photograph and a detailed block diagram of the EPS I module, respectively. Further details regarding each connector labeled in Figure 1 and the corresponding pinouts can be found in Section 3.2.1.

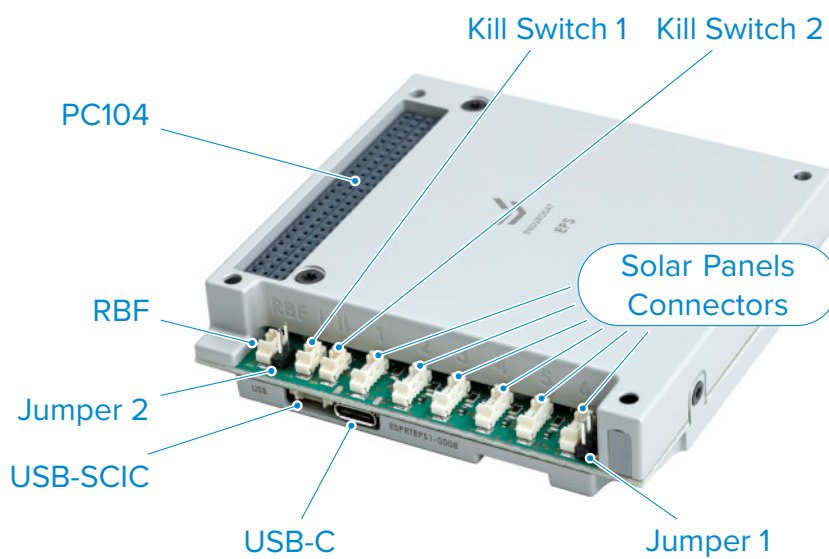


Figure 1: EPS I module



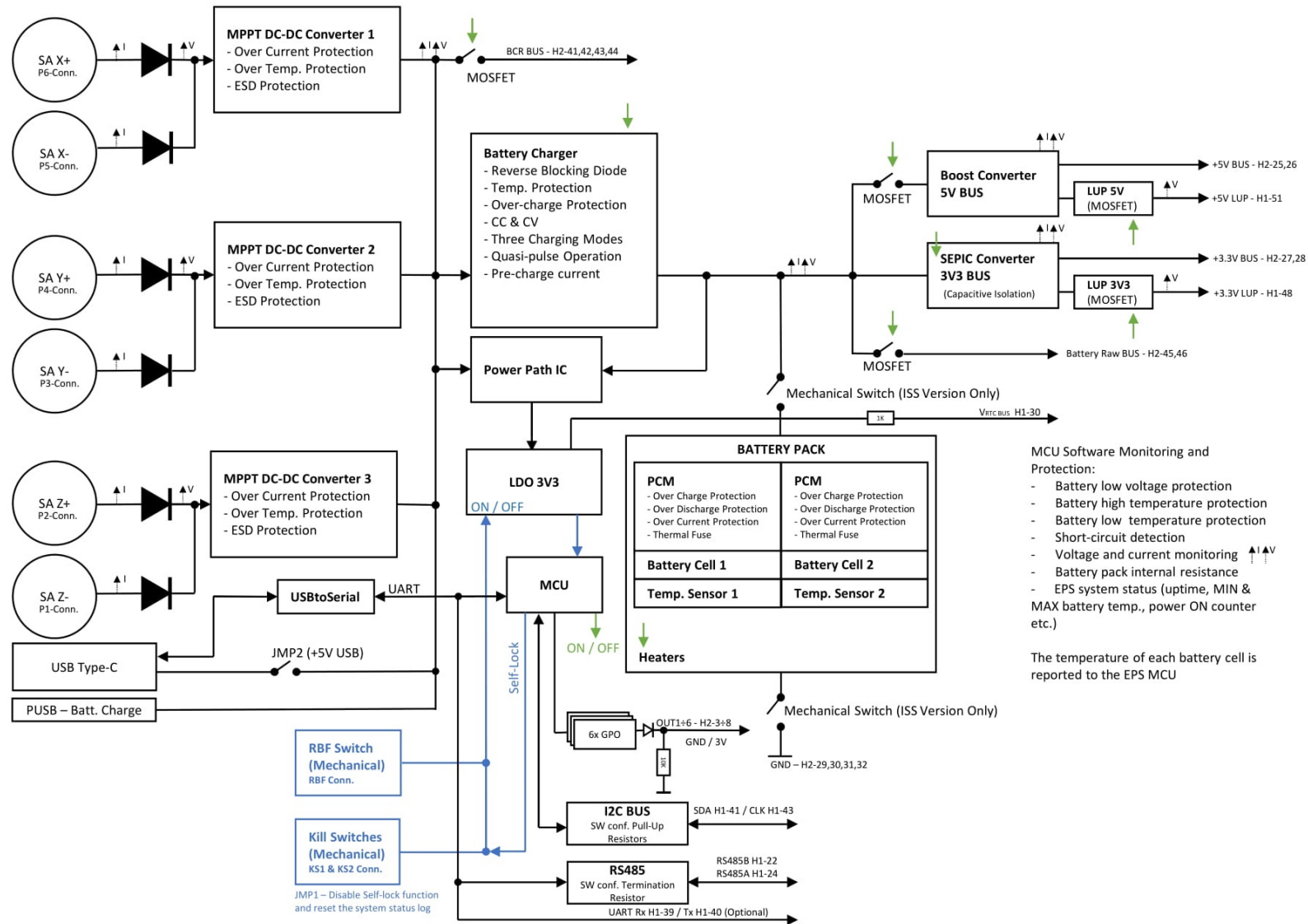


Figure 2: Block diagram of the EPS I Module



### EPS I vs. EPS I Plus

EnduroSat offers two variants of the module - the EPS I and the EPS I Plus. They both comprise of identical hardware and make use of Lithium Polymer (Li-Po) batteries. The difference is that the EPS I module includes a single battery pack consisting of two batteries, while the EPS I Plus module includes two such battery packs in parallel. This means that the EPS I Plus module is heavier, thicker, packs twice the battery capacity, and can provide twice the maximum cumulative output power when compared to its EPS I counterpart (Table 2).

Table 2: Comparison of the EPS I and the EPS I Plus modules

| Parameter                       | Unit | EPS I | EPS I Plus |
|---------------------------------|------|-------|------------|
| Battery Capacity                | Wh   | 10.2  | 20.4       |
| Maximum Cumulative Output Power | W    | 10    | 20         |
| Mass                            | g    | 229.5 | 329.7      |
| Thickness                       | mm   | 21.2  | 30.0       |



**Note:** For the sake of brevity, throughout this manual the term EPS I is understood to mean both the EPS I and EPS I Plus modules, unless explicitly stated otherwise.

### Solar Panel Channels

EPS I has three Solar Panel Channels (denoted by the three Maximum Power Point Tracking (MPPT) DC-DC converter blocks in Fig. 2) and six solar panel connectors (one connector per CubeSat sides, illustrated in Fig. 1 and on denoted by the blocks on the very left of Fig. 2). Each Solar Panel Channel is fed by two solar panel connectors, which in turn should be connected to Solar Panels (SPs) mounted on two opposite sides of the CubeSat.

The three Solar Panel Channels are connected in parallel and can operate at the same time. Thus, the total generated energy is effectively the sum of the energy generated by all solar panels.

The output energy for each solar panel is monitored since each Solar Panel Channel has a DC-DC step-up converter with MPPT. Particularly, the step-up converter turns OFF, when the input voltage is less than  $V_{Fblockingdiode} + 0.27\text{ V}$  (under voltage lockout) and restarts when the voltage exceeds  $V_{Fblockingdiode} + 0.34\text{ V}$ . A hysteresis is implemented to avoid unpredictable ON-OFF switching. However, the minimum input voltage threshold for boost is  $V_{Fblockingdiode} + 0.5\text{ V}$  (under voltage lockout). A hysteresis is implemented to avoid an unpredictable ON-OFF switching. Please refer to Table 12.



Each Solar Panel Channel has the following features:

- Accurate current and voltage measurement;
- Over-current protection – engages above 1.8 A;
- Over-voltage protection – engages above 5.5 V;
- Over-temperature protection
- ESD protection;
- High boost converters efficiency – up to 95%.

### **Battery Charger**

The battery charger operates in both linear and quasi-pulse modes and ensures Constant Current - Constant Voltage (CC-CV) charging of the batteries. The quasi-pulse charging mode has the advantage of allowing the energy from weakly illuminated solar cells to be very efficiently harvested. In addition, users can select among three different battery charging current levels, which ensures a good match with the power generated by a variety of solar panel configurations. For further details on the available charging current levels, please consult Table 12. The efficiency of the battery charger is approximately 90%.

### **Protection Circuit Module (PCM)**

Each battery cell incorporates hardware protection against over-current, over-charge and over-discharge. If the PCMs are triggered (i.e. the battery cells become disconnected from the EPS I), then all power bus outputs from the EPS are turned off. In this case the following occurs:

- Over-current: the PCM automatically resets and reconnects the battery pack to the EPS.
- Over-charge: the PCM automatically reset when the charging voltage drops below 4.275 V.
- Over-discharge: the PCM automatically reset when the battery cell voltage goes above 2.3 V.



## Protection Algorithm

As an additional safety feature, EnduroSat has developed a firmware algorithm to protect the batteries from short circuit, deep discharge and overheating (Fig. 3).

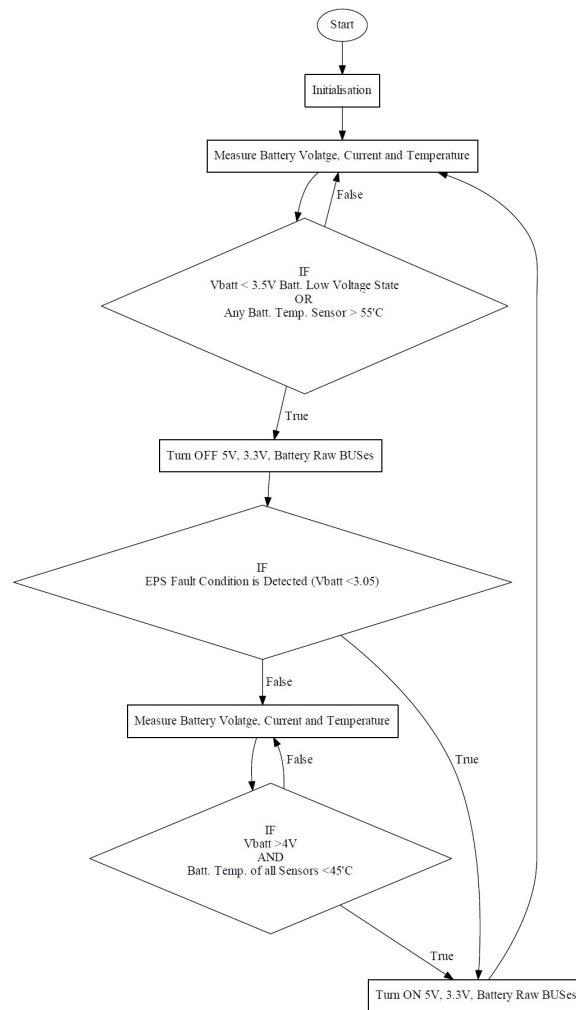


Figure 3: Algorithm for protection of the batteries from short circuit, deep discharge and overheating.



## Battery Heaters

Each battery pack has three independent heaters to prevent battery charging below 0 °C. The EPS I incorporates a special algorithm which optimizes the heater power consumption as a function of temperature (Fig. 4).

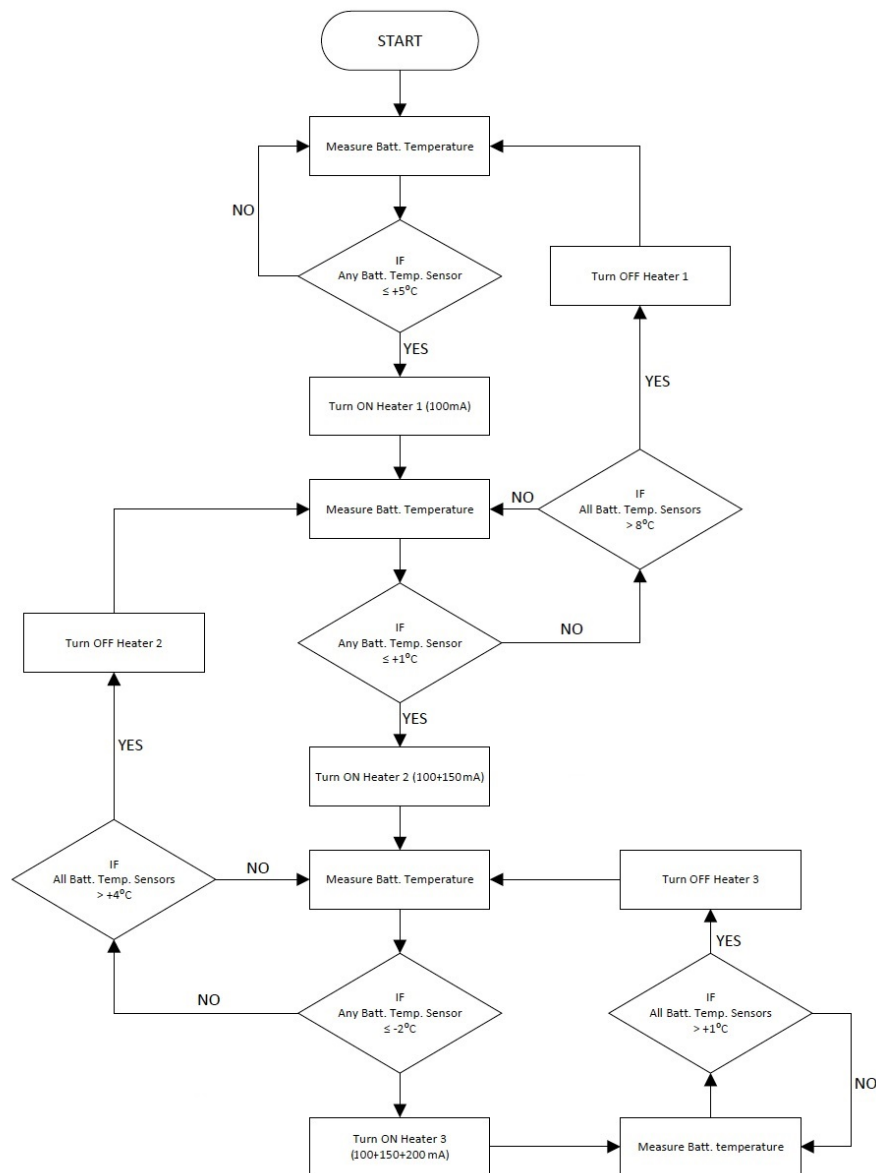


Figure 4: Algorithm for optimization of heater power consumption.



### 3.1 Mechanical Description

The mechanical drawings of EnduroSat's EPS I are shown in Figs. 5 and 6 below. All dimensions are in mm.



**Warning:** Make sure to use threaded bars M3 or equivalent diameter bars through the mounting apertures of the module to ensure proper fit during assembly.

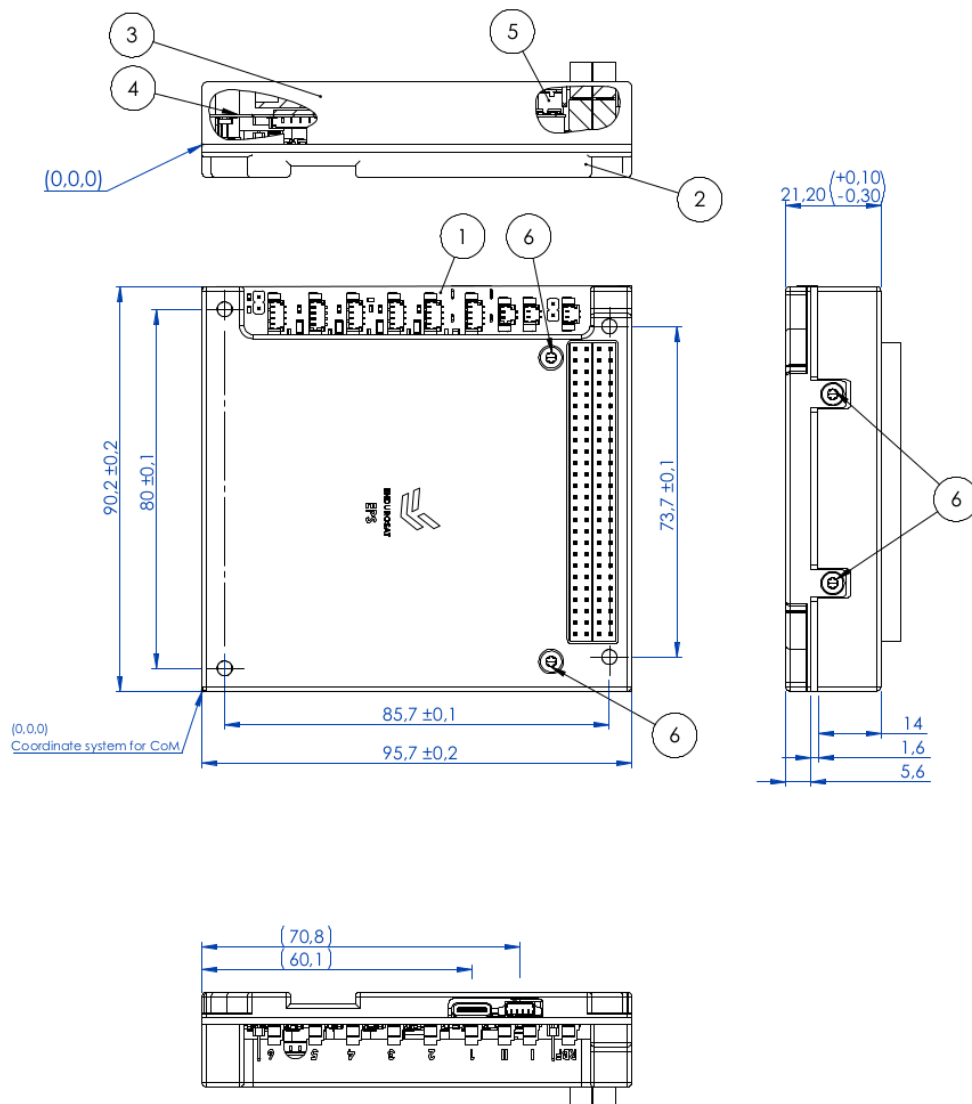


Figure 5: EPS I (includes 1 battery pack)

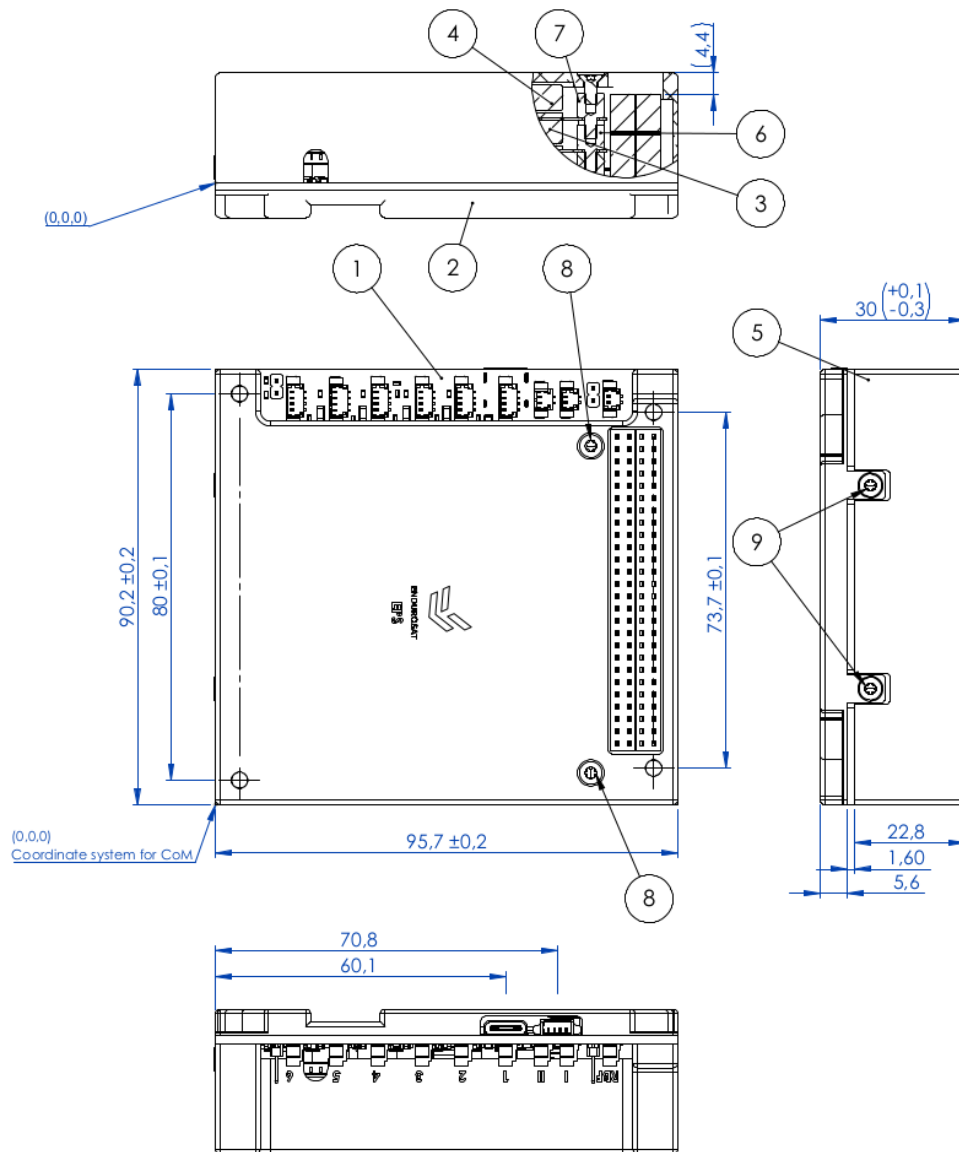


Figure 6: EPS I Plus (includes 2 battery packs)



## 3.2 Connectivity

### 3.2.1 Connectors & Pinouts



Figure 7: EPS I module top view

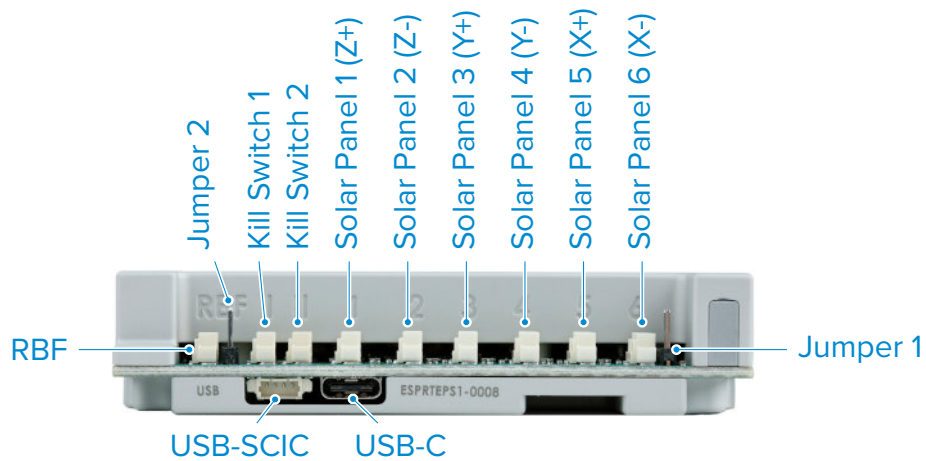


Figure 8: EPS I module side view





## H1 & H2-PC/104 Stack Connector

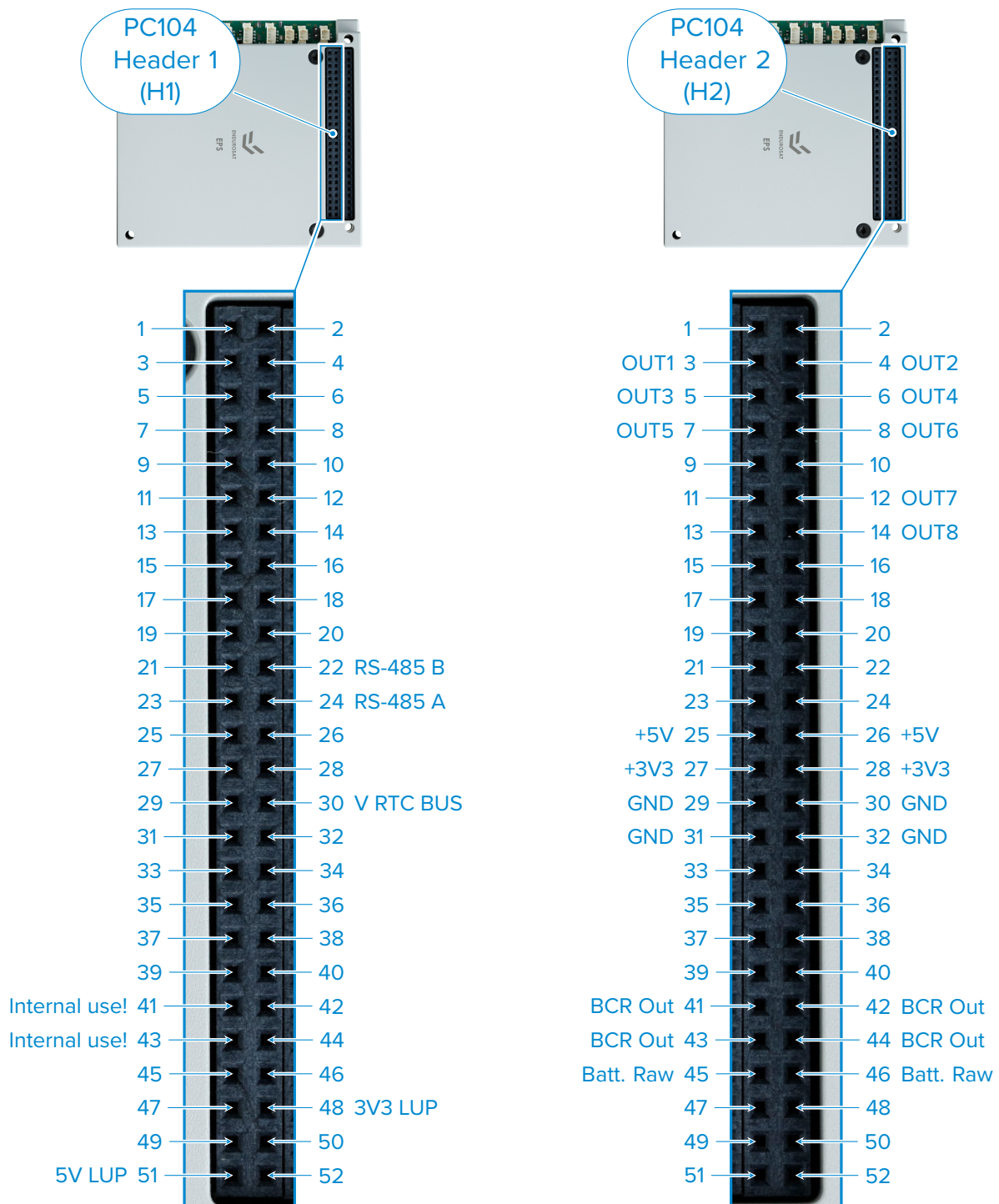


Figure 9: EPS PC104 pinout diagram



Table 3: Pinouts of H1 connector

| Pin   | Mnemonic                 | Description                                          |
|-------|--------------------------|------------------------------------------------------|
| H1-22 | RS485B <sup>1</sup>      | Driver output or receiver input (complementary to A) |
| H1-24 | RS485A <sup>1</sup>      | Driver output or receiver input (complementary to B) |
| H1-30 | V RTC BUS                | Supply voltage for RTC of external device            |
| H1-41 | Sys I <sup>2</sup> C SDA | Reserved for internal use!                           |
| H1-43 | Sys I <sup>2</sup> C SCL | Reserved for internal use!                           |
| H1-48 | 3V3 LUP                  | +3.3V BUS                                            |
| H1-51 | 5V LUP                   | +5V BUS                                              |

<sup>1</sup> Can be used with ESPS I protocol



Table 4: Pinouts of H2 connector

| Pin   | Mnemonic  | Description                       |
|-------|-----------|-----------------------------------|
| H2-3  | OUT1      | Universal Output 1                |
| H2-4  | OUT2      | Universal Output 2                |
| H2-5  | OUT3      | Universal Output 3                |
| H2-6  | OUT4      | Universal Output 4                |
| H2-7  | OUT5      | Universal Output 5                |
| H2-8  | OUT6      | Universal Output 6                |
| H2-12 | OUT7      | Universal Output 7                |
| H2-14 | OUT8      | Universal Output 8                |
| H2-25 | +5V       | +5V BUS                           |
| H2-26 | +5V       | +5V BUS                           |
| H2-27 | 3V3       | +3.3V BUS                         |
| H2-28 | 3V3       | +3.3V BUS                         |
| H2-29 | GND       | Ground                            |
| H2-30 | GND       | Ground                            |
| H2-31 | GND       | Ground                            |
| H2-32 | GND       | Ground                            |
| H2-41 | BCR Out   | Battery Charger Regulator Voltage |
| H2-42 | BCR Out   | Battery Charger Regulator Voltage |
| H2-43 | BCR Out   | Battery Charger Regulator Voltage |
| H2-44 | BCR Out   | Battery Charger Regulator Voltage |
| H2-45 | Batt. Raw | Battery voltage                   |
| H2-46 | Batt. Raw | Battery voltage                   |



## USB Connectors

- USB Type-C

Can be used for monitoring and controlling the EPS module. When the JMP2 jumper is inserted the internal batteries can be charged through this connector.

- PUSB

Secondary USB connector, which serves for connecting the EPS module to the solar panel SCIC connector in order to charge the batteries and provide access to the USB interface when the satellite is fully assembled.

Powered Universal Serial Bus (PUSB) is connected electrically in parallel to the USB Type-C. The purpose of this connector is to enable the user to access the EPS module when the satellite is fully assembled.

At the time that the satellite is fully assembled, the USB Type-C and PUSB connectors may not be physically accessible. Therefore, a custom USB cable can be connected from the PUSB connector to the SCIC on the bottom side of EnduroSat's X/Y RBF Solar Panel. The SCIC of the CubeSat is also accessible from the top side of the Solar Panel and this allows the user to easily configure and monitor the EPS I module as well as to charge the batteries.

Table 5: Pinouts of PUSB Connector.

| Pin | Mnemonic | Description     |
|-----|----------|-----------------|
| 1   | +5V      | +5Vdc USB Input |
| 2   | USBN     | Data Negative   |
| 3   | USBP     | Data Positive   |
| 4   | GRDN     | Ground          |

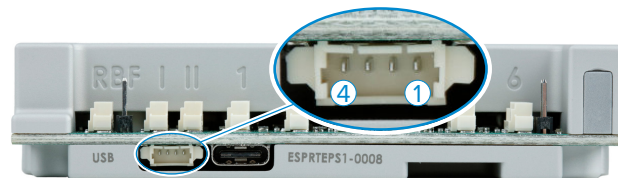


Figure 10: USB-SCIC connector.

The connector pins are numbered sequentially in increasing order from right to left. The positions of the 1<sup>st</sup> pin and 4<sup>th</sup> (last) pin are indicated on the magnified area on Figure 10.



Table 6: PUBS Connector Specifications

| Connector Name | Part Number | Trade Name | Mates with |
|----------------|-------------|------------|------------|
| PUSB Connector | 53261-0471  | Pico-Blade | 51021-0400 |

### P1 to P6 - EPS I Solar Panel Connectors

The EPS module has six solar panel connectors from P1 to P6, and they are 4-pin MOLEX Pico-Blade 53398-0471.



Figure 11: EPS Solar Panel connector.

The connector to Solar Panel 1 (Z+) is shown, however all other five Solar Panel connectors are identical. The connector pins are numbered sequentially in increasing order from left to right. The positions of the 1<sup>st</sup> pin and 4<sup>th</sup> (last) pin are indicated on the magnified area.



EnduroSat's recommended configuration is as follows:

Table 7: EPS I Solar Panel Connectors and Associated Axis.

|    |    |
|----|----|
| P1 | Z+ |
| P2 | Z- |
| P3 | Y+ |
| P4 | Y- |
| P5 | X+ |
| P6 | X- |

This configuration can be modified, taking into account that the pairs: P1 and P2, P3 and P4, P5 and P6 should be connected to the solar panels on opposite sides of the same axis.

The pinouts for the EPS I Solar Panel Connector are the same for each of the 6 connectors (P1 - P6):

Table 8: Pinouts of each EPS I Solar Panel Connector.

| Pin | Mnemonic | Description |
|-----|----------|-------------|
| 1   | -        | Negative    |
| 2   | -        | Negative    |
| 3   | +        | Positive    |
| 4   | +        | Positive    |

Table 9: EPS I Solar Panel Connectors Specifications

| Connector Name | Part Number | Trade Name | Mates with |
|----------------|-------------|------------|------------|
| P1             | 53398-0471  | Pico-Blade | 51021-0400 |
| P2             | 53398-0471  | Pico-Blade | 51021-0400 |
| P3             | 53398-0471  | Pico-Blade | 51021-0400 |
| P4             | 53398-0471  | Pico-Blade | 51021-0400 |
| P5             | 53398-0471  | Pico-Blade | 51021-0400 |
| P6             | 53398-0471  | Pico-Blade | 51021-0400 |



RBF Connector



Figure 12: RBF Connector

Table 10: RBF Connector Specifications

| Connector Name | Part Number | Trade Name | Mates with |
|----------------|-------------|------------|------------|
| RBF            | 53398-0271  | Pico-Blade | 51021-0200 |

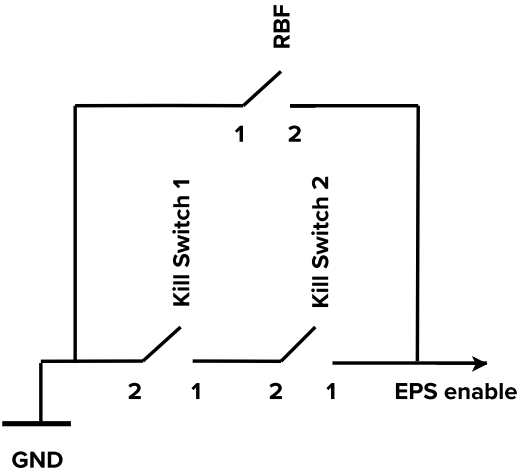


Figure 13: RBF and Kill Switches pins



KILL Switches Connectors



Figure 14: Kill Switches Connectors

Table 11: Kill Switches Specifications

| Connector Name | Part Number | Trade Name | Mates with |
|----------------|-------------|------------|------------|
| Kill Switch 1  | 53398-0271  | Pico-Blade | 51021-0200 |
| Kill Switch 2  | 53398-0271  | Pico-Blade | 51021-0200 |





## 4 System Functionality

### 4.1 Electrical Characteristics

Table 12: Electrical Characteristics

| Parameter                     | Unit | Condition                                 | Min.                  | Typical | Max.             |
|-------------------------------|------|-------------------------------------------|-----------------------|---------|------------------|
| BATTERY CAPACITY              |      |                                           |                       |         |                  |
| Capacity                      | mAh  | EPS with 1 battery pack                   | 2640                  | 2800    |                  |
|                               | mAh  | EPS with 2 battery packs                  | 5280                  | 5600    |                  |
|                               | Wh   | EPS with 1 battery pack                   |                       | 10.4    |                  |
|                               | Wh   | EPS with 2 battery packs                  |                       | 20.8    |                  |
| BATTERY CHARGER               |      |                                           |                       |         |                  |
| EOC Voltage                   | V    |                                           | 4.16                  | 4.2     | 4.24             |
| Charging Current              | mA   | Charge Mode 3                             | 640                   | 690     | 740              |
|                               |      | Charge Mode 2 (default)                   | 430                   | 460     | 490              |
|                               |      | Charge Mode 1                             | 215                   | 230     | 245              |
| BATTERY DISCHARGE             |      |                                           |                       |         |                  |
| Overdischarge Detection       | V    | Limited by PCM                            | 2.24                  | 2.3     | 2.36             |
| Overcurrent Detection         | mA   | EPS with 1 battery pack (limited by PCM)  |                       | 4000    | 9000<br>8–16 ms  |
|                               | mA   | EPS with 2 battery packs (limited by PCM) |                       | 8000    | 16000<br>8–16 ms |
| Expected # of Cycles/Lifetime | mAh  | Discharge @ 0.5/1°C & 23±2 °C             | 500 cycles ≥ 2196 mAh |         |                  |
| UNREGULATED BATTERY BUS       |      |                                           |                       |         |                  |
| Output Voltage                | V    | Firmware defined                          | 3.5                   |         | 4.12             |

Continued on the following page



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| Parameter                 | Unit | Condition                                                          | Min. | Typical | Max.             |
|---------------------------|------|--------------------------------------------------------------------|------|---------|------------------|
| Output Current            | mA   | EPS with 1 battery pack<br>(limited by PCM)                        |      | 4000    | 9000<br>8–16 ms  |
|                           | mA   | EPS with 2 battery packs<br>(limited by PCM)                       |      | 8000    | 16000<br>8–16 ms |
| <b>+5 V BUS</b>           |      |                                                                    |      |         |                  |
| Output Voltage            | V    |                                                                    | 4.88 | 5       | 5.15             |
| Output Current            | mA   | EPS with 1 battery pack                                            |      |         | 2000             |
|                           |      | EPS with 2 battery packs                                           |      |         | 4000             |
| Operating Frequency       | kHz  |                                                                    | 500  | 535     | 560              |
| Efficiency                |      | $V_{\text{batt}} = 4 \text{ V} ; I_{5\text{VBUS}} = 2 \text{ A}$   | 82%  | 84%     | 86%              |
| <b>+3.3 V BUS</b>         |      |                                                                    |      |         |                  |
| Output Voltage            | V    |                                                                    | 3.3  | 3.38    | 3.45             |
| Output Current            | mA   | EPS with 1 battery pack                                            |      |         | 3000             |
|                           |      | EPS with 2 battery packs                                           |      |         | 4000             |
| Operating Frequency       | kHz  |                                                                    | 465  | 495     | 525              |
| Efficiency                |      | $V_{\text{batt}} = 4 \text{ V} ; I_{3.3\text{VBUS}} = 2 \text{ A}$ | 72%  | 76%     | 78%              |
| <b>MODULE CONSUMPTION</b> |      |                                                                    |      |         |                  |
| Current Consumption       | mA   | Batt. Low State @ 3.3 V                                            |      | 13      |                  |
|                           |      | Normal Operation @ 3.7 V                                           |      | 20      |                  |
|                           |      | LUP3V3 & LUP5V                                                     | 3.9  |         | 5                |
|                           |      | Heater 1 -1×Battery Pack                                           | 90   | 100     | 115              |
|                           |      | Heater 2 -1×Battery Pack                                           | 140  | 150     | 170              |
|                           |      | Heater 3 -1×Battery Pack                                           | 185  | 200     | 225              |
|                           |      | Heater 1 -2×Battery Pack                                           | 180  | 200     | 230              |

Continued on the following page



Continued from the previous page

| Parameter                                        | Unit | Condition                                                | Min. | Typical | Max. |
|--------------------------------------------------|------|----------------------------------------------------------|------|---------|------|
| Power Consumption                                | mW   | Heater 2 -2×Battery Pack                                 | 280  | 300     | 340  |
|                                                  |      | Heater 3 -2×Battery Pack                                 | 370  | 400     | 450  |
|                                                  |      | Normal operation; LUP5V & LUP3V3 are OFF                 |      | 75      |      |
|                                                  |      | Low voltage or high temperature state; all buses are OFF |      | 43      |      |
|                                                  |      |                                                          |      |         |      |
| Fault Current Threshold (auto retry every 15 ms) | A    | LUP3V3 & LUP5V                                           | 1.75 |         | 2.2  |

#### SOLAR PANELS INPUT

|                                           |    |                 |   |       |      |
|-------------------------------------------|----|-----------------|---|-------|------|
| Voltage                                   | V  | @ 25 °C         | 0 | 4.66  | 5.5  |
| Current                                   | mA | @ 25 °C         |   | 517   | 1800 |
| Power                                     | W  | @ 25 °C         |   | 2.6   |      |
| Integrated Blocking Diode Forward Voltage | V  | @ 1A & 25 °C    |   | 0.325 |      |
|                                           |    | @ 0.5 A & 25 °C |   | 0.275 |      |

The CubeSat has two deployment switches located at the two opposite corners of the Z-face of the CubeSat structure, in accordance with the CubeSat design specification. These two deployment switches should be connected to the dedicated deployment connectors on the EPS module. When the satellite is in space, there is no connection between the two pins of each deployment switch connector. When the satellite is in the deployment pod, there is a short connection between the two pins of each deployment switch connector.

There is also a RBF located on the side of the CubeSat structure, in accordance with the CubeSat design specification. The RBF switch should be connected to the dedicated RBF connector on the EPS module.

When the RBF pin is plugged in and both deployment switches are pressed, the EPS module is inactive. In order to activate the EPS module, it is necessary that the RBF pin is unplugged and



at least one deployment switch is released. The deployment switches and the RBF switch are peripheral circuits to the EPS module. The internal Microcontroller Unit (MCU) controls the high impedance internal MOSFET switches in the EPS, depending on the state of the deployment and RBF switches. The MOSFET switches connect/disconnect the internal battery pack(s) from the EPS module.

Once the EPS module is turned on, then the self-locking functionality is activated, and the power supply cannot be stopped by resetting the deployment switches or the RBF pin. For ground test purposes, the self-lock functionality can be stopped using the PC software, via the USB port or from the dedicated jumper JMP1. In addition to disabling self-lock, JMP1 also resets the system counters (Power On cycles, Min & Max Temperature etc).

The incoming power along with the energy stored in the batteries is used to feed all the main buses: 3.3V@3A, 5V@2A and 'battery raw'@4A (specifications are for EPS I with 1 Battery Pack).

There are two LUP outputs: one internally connected to the 5V BUS (H1-48) and the other one internally connected to the 3.3V BUS (H1-51). The trip current threshold is 2 A for both LUP outputs. If the trip current threshold is reached the LUP output is disconnected. An auto retry mechanism is implemented and every 15ms the LUP outputs check if the current is below the trip threshold for normal operation to continue. This trip condition can be reported to an external module such as the On-board Computer (OBC).

When the EPS is turned on, a special 3.3V output power bus (Voltage Output for Real Time Clock (VRTC) BUS) from the EPS, supplies an external RTC in EnduroSat's OBC. This VRTC BUS cannot be shut off even when a low battery state occurs. The current on this VRTC BUS is limited with a 1k $\Omega$  serial resistor. Please note, when the satellite is turned off and ready for deployment in the P-POD then the VRTC BUS is also off. This RTC is also off and must be set in orbit after the satellite has been deployed.

## 4.2 EPS Module Monitoring

There is built-in firmware in the EPS module for monitoring, which can report (via the external interfaces) the following parameters:

- ✓ Voltage and current generated by each solar panel. N.B. the voltage measurement is made after the solar panel blocking diode.
- ✓ Combined voltage and current of all three Solar Panel Channels.
- ✓ Output current and voltage of each power bus (5V BUS, 3.3V BUS, Battery BUS and BCR BUS).



- ✓ Output voltage of LUP3.3V & LUP5V.
- ✓ Battery pack(s) voltage, current and temperature (one temperature sensor per battery cell).
- ✓ Up to 4 main (for batteries) and 3 additional external temperature sensors.
- ✓ Critical fail status: power cycle, low voltage, EPS fault conditions, over temperature, and minimum and maximum temperatures reached for each battery cell. All the outputs are switched OFF (deactivated) ASAP.
- ✓ On/Off status of the output power buses.
- ✓ On/Off status of all six GPIO of the EPS module
- ✓ Charge mode status.
- ✓ Self-lock function status.
- ✓ Battery heater status.

The EPS module measures the current from each solar panel, and the Solar Panel Channel voltage. The total output voltage of the three Solar Panel Channels is also monitored. Each Solar Panel Channel has a step-up converter with MPPT.

The EPS module measures the output current and voltage of all the power buses: 5V, 3.3V, 'battery raw' and BCR Out.

The EPS module monitors the instantaneous temperature of every battery cell. It also monitors the minimum and maximum temperature of every battery cell (after they have been reset).

On the first battery pack there is a single connector for an additional three external temperature sensors. The temperature sensors can be mounted anywhere with the satellite.

### 4.3 EPS Module Control

The firmware provides control (via the external interfaces) of the following features:

- ✓ On/Off state of the self-lock function
- ✓ On/Off state of the power BUS: BCR, 3.3V, 5V and 'battery raw'
- ✓ On/Off state of the 3.3V and 5V LUP outputs
- ✓ Set the battery charge mode (3 different options/currents)
- ✓ On/Off state of the battery heaters



✓ On/Off state of the six General-Purpose Input/Outputs (GPIOs)

Particularly, the EPS module has six GPIOs (General Purpose Outputs). Every output can be switched between 3V and Ground (GND). All the outputs are protected with diodes and 10k pull-down resistors, and this enables other modules to control them at the same time. Diode or gates can be realized on these outputs.

Some outputs are activated with high level, and others with low level. At “high/low level” represents the appropriate pin state, and “active/inactive” - the appropriate state which depends on “active low/active high” for the appropriate output. Additionally, it is not possible to be sure in terms of the activation polarity for OUT\_1 ... OUT\_8. This is fully dependent on the Customer’s set up.

Concretely, the outputs are divided into 3 types, depends on possible current consumption:

- Priority outputs - almost no power consumption
- Normal
- High-current consumption - outputs with possible significant current flow on its activation.

The desired output state will be set immediately, while actual output state could be changed with some delay, as explained below.

Every 1ms (occasionally could be more due to performed tasks), for each OUTPUT/BUS channel, the following sequence is performed:

1. All the “Priority” outputs are switched to desired state;
2. Only one of “Normal” outputs is switched to desired state;
3. Only one of “High-current” outputs is switched ON, with additional condition: during last 20ms none of “High-current” switches was turned ON.

So, looking into the sequence, even if several high-current switches were turned ON at the same time, the desired state of outputs could be reached with significant delay ( $[ N \times 20\text{ms} ] + 1$ ) ms, where N is the high-current outputs counter.

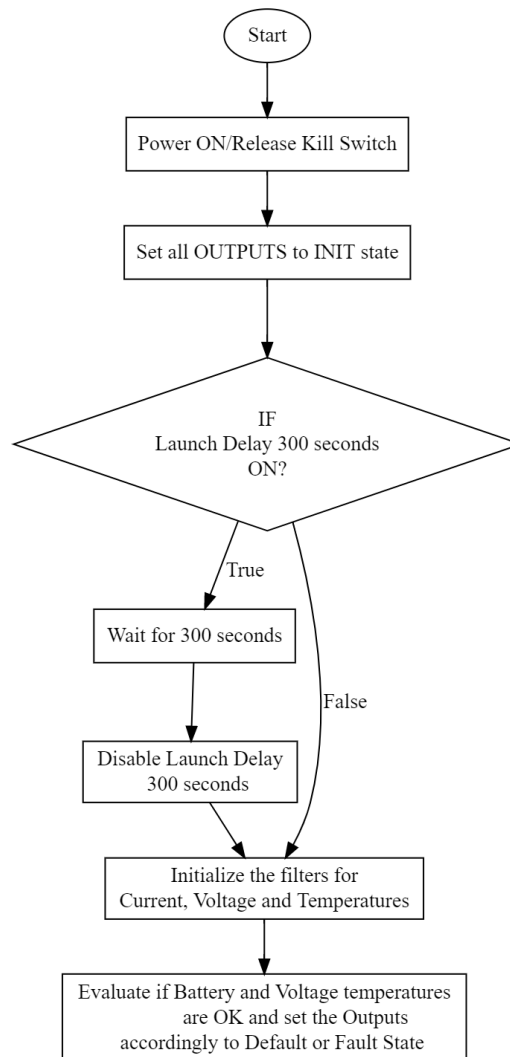


Figure 15: Algorithm for the device power-on sequence



**Note:** The algorithm is valid only when the EPS is integrated in a platform, not as a standalone product. The launch delay period (that is 300 sec) can be changed by the 0x0A queryLaunchState command.



## 4.4 Firmware Update

It is recommended that the firmware is changed in a lab rather than in space. The Bootloader does not support receiving commands from the ground. If the EPS works only in Bootloader mode and a failure to flash the FW application arises (not likely to happen in low radiation environment), then the EPS does not ensure protection against too low/too high temperatures. Provided that a new FW upgrade is available, EnduroSat provides the tool, the update file and the corresponding description upon request. The FW upgrade requires only a USB-C cable. The procedure can be repeated multiple times if necessary (due to switching to previous revision for a while or other reason).





## 5 System Software

### 5.1 ESPS I

EnduroSat's EPS I and EPS I Plus module are loaded with EnduroSat Protocol Stack I (ESPS I) firmware. The ESPS I is a complete software communication framework specifically designed by EnduroSat for nanosatellites. It consists of several protocols, each one of which carries out specific tasks within a single or multiple network layers. These protocols can be stacked together in various ways in order to accommodate a number of different use case scenarios.

Broadly speaking, the functionality of ESPS I can be conceptually subdivided into two general categories. ESPS I enables:

- Inter-module communication within a CubeSat
- Communication between a GS and CubeSat modules

More specifically, the ESPS I allows users to read telemetry and run commands in the EPS I and EPS I Plus modules. Furthermore, it also allows users to relay commands to additional modules within the spacecraft (if integrated under the same protocol) and retrieve telemetry logs. For further information in the structure and composition of this protocol stack, please refer to ***"EnduroSat's ESPS I User Manual"***.

The following sections provide a breakdown of all the available interfaces within the EPS I and EPS I Plus modules. As well as the structure and data types required for inputs and output arguments.



**Note:** The information outlined in the following sections is intended to depict the general capabilities of the EPS in terms of collecting housekeeping telemetry, controlling power lines, and general information about the module.

For a full understanding of a complete implementation of this protocol with ground support equipment and in-the-air operations, the reader shall refer to ***"EnduroSat's Space Development Environment"*** and ***"EnduroSat's Digital Ground Station"*** documentation.

EnduroSat is able to provide STM32-ready reference implementation of ESPS I to customers for the microcontroller. Through this reference documentation and implementations, users may integrate payloads and additional modules into the ESPS I stack and communicate in the same approach as EnduroSat core subsystems.



## 5.2 Module Interfaces

The EPS I module has the capability of sustaining nominal operations in both roles as a Master and Slave. The following interfaces described in Table 13 are defined for each of these roles, which depend on specific conditions. It shall be noted that the following module interfaces are valid for EnduroSat Power module EPS type I , Hardware versions  $\geq 1.7$ .

Table 13: Module Interfaces

| Interface Name | Section | Description                                                                                                                                                                               | Deployed as |
|----------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| EPS I          | 6.1     | These interface methods correspond to the client requests for telemetry and set parameters as described.                                                                                  | Client      |
| EPS I Notify   | 6.4     | These interface methods correspond to server notifications to all modules. The master behavior is inherited upon State Machine conditions where survivability of the system is preserved. | Server      |

### General notes:

Commands/methods that return telemetry data such as temperature, voltage, current, and other physical parameters adopt the following calibration approach:

$$\text{Physical Quantity} = \text{Raw Value} * \text{Calibration Factor}$$



## 6 Software Commands

### 6.1 EPS I Interface Methods Overview

EnduroSat Power module type I, HW version  $\geq 1.7$ . The following interface methods correspond to the client requests for telemetry and set parameters as described hereby.

| Method name          | Method description                                                                                           |
|----------------------|--------------------------------------------------------------------------------------------------------------|
| getSoftwareVersion   | Acquire the software version information for the EPS 1 module.                                               |
| getBatteryInfo       | Reads the battery information and telemetry of the EPS 1 Module.                                             |
| getBusesInfo         | Reads the buses information and telemetry of the EPS 1 module.                                               |
| getSolarPanelsInfo   | Provides solar panel information and telemetry (voltage/current) parameters as received by the EPS 1 module. |
| getStatisticInfo     | Reads statistic data                                                                                         |
| getTemperaturesInfo  | Reads the temperatures of MCU and batteries                                                                  |
| getConfigurationInfo | Reads the current state of inputs and outputs, and its default values                                        |
| queryControls        | Changes the outputs state and receives its current states                                                    |
| queryRBatRaw         | Sets the initial value of the resistance of the battery                                                      |
| queryDefaults        | Changes the default outputs state and receives its current default states after executing the command        |
| queryLaunchState     | Changes the default outputs state and receives its current states                                            |
| queryWatchdog        | Get/Set watchdog configuration - timer and output mask to reset                                              |
| resetWatchdog        | Resets the Watchdog timer in order to prevent MCU Reset on its timeout                                       |

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| Method name            | Method description                                                                                                                                                                                                   |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resetPowerSupply       | Resets the MCU and/or all the outputs the delay of 100ms before reset is implemented in order to confirm the command. however, on a high ESPS load it is possible that response will not be sent due to time limits. |
| SetSerialNumber        | Write (once per lifetime) and Read afterwards the EPS1 serial number                                                                                                                                                 |
| GetSerialNumber        | Read the EPS1 serial number                                                                                                                                                                                          |
| HIBModeRead            | Read the Settings of HIB mode                                                                                                                                                                                        |
| HIBModeWrite           | Write the Settings of HIB mode                                                                                                                                                                                       |
| HeatersSettingsRead    | Read the Settings of all three heaters                                                                                                                                                                               |
| HeatersSettingsWrite   | Write and Read afterwards the Settings of all heaters                                                                                                                                                                |
| ToggleOutPut           | Changed the state of one output and after the toggle timeout will set it back to its original state                                                                                                                  |
| NotificationsSettings  | The EPS I will send broadcast commands to notify other devices for an event that may result in power down of all payloads.                                                                                           |
| GetTelemetryAltogether | some of the previous commands combined                                                                                                                                                                               |
| bload_getVersion       | Gets the current Bootloader version, if there is any                                                                                                                                                                 |
| bload_eraseFlash       | Command to erase the storage flash memory for writing the new image of the application                                                                                                                               |
| bload_writeFlash       | Writes part of a image file of a new application in the storage flash memory                                                                                                                                         |
| bload_execute          | Request the installation of the new image from the storage flash over the current application                                                                                                                        |



## 6.2 Data Types

### VersionVariant

Helper from getSoftwareVersion, version variant 0xFF for development state 0x00 for release candidate 1..100 for major version number 101-254 reserved

### Literals

| Literal name     | Literal value | Description       |
|------------------|---------------|-------------------|
| DevelopmentState | 255           | Development State |
| ReleaseCandidate | 0             | Release Candidate |
| Release          | 1             | Release Ready     |

### VA\_Pair

Helper structure to hold pair voltage/current. The returned values correspond to raw values and shall be calibrated by the user through the following calibration coefficients and formulas (when applicable).

### Attributes

| Attribute name | Attribute type | Description                                                                                                                                                                                                                                     |
|----------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volt           | UInt16         | Calibration Batt: *0.0023394775 [V]<br>Calibration BCR: *0.0023394775 [V]<br>Calibration 3V3 BUS: *0.0023394775 [V]<br>Calibration 5V BUS: *0.0023394775 [V]<br>Calibration LUP 3V3: *0.0023394775 [V]<br>Calibration LUP 5V: *0.0023394775 [V] |
| curr           | UInt16         | Calibration Batt: *0.0030517578 [A]<br>Calibration BCR: *0.0015258789 [A]<br>Calibration 3V3 BUS: *0.0020345052 [A]<br>Calibration 5V BUS: *0.0020345052 [A]                                                                                    |



## VAA\_Pack

Helper structure to keep voltage, negative and positive current. The returned values correspond to raw values and shall be calibrated by the user through the following calibration coefficients and formulas (when applicable).

### Attributes

| Attribute name | Attribute type | Description                    |
|----------------|----------------|--------------------------------|
| volt           | UInt16         | Calibration: *0.0024414063 [V] |
| curr_neg       | UInt16         | Calibration: *0.0006103516 [A] |
| curr_pos       | UInt16         | Calibration: *0.0006103516 [A] |

### Statistic\_Counters

Helper structure to keep counters for "getStatisticInfo"

### Attributes

| Attribute name  | Attribute type | Description                          |
|-----------------|----------------|--------------------------------------|
| powerOn         | UInt32         | Counter of incidence. No calibration |
| chargeCycle     | UInt32         | Counter of incidence. No calibration |
| underVoltage    | UInt32         | Counter of incidence. No calibration |
| shortcutCircuit | UInt32         | Counter of incidence. No calibration |
| overtemperature | UInt32         | Counter of incidence. No calibration |



## QueryState

Helper description from "queryControls" and "queryDefaults"

### Literals

| Literal name | Literal value | Description                                                   |
|--------------|---------------|---------------------------------------------------------------|
| Ignored      | 0             | Ignore - does not change, if used return the actual value     |
| Off          | 1             | Toggle OFF the specific control, as output, returns if OFF    |
| On           | 2             | Toggle ON the specific control, as output, return if ON       |
| Default      | 3             | Returns to Default value                                      |
| ForceOff     | 5             | Force the control to OFF                                      |
| ForceOn      | 6             | Force the control to ON                                       |
| ValueMask    | 15            | Mask for toggling values, above                               |
| FlagsMask    | 192           | Mask for toggling flags below                                 |
| Reserved     | 64            | Reserved, not used                                            |
| Mismatch     | 128           | Mismatch between what was asked and what is the current state |

## QueryControlsIndex

Helper enum for accessing "queryControls" elements by named index

### Literals

| Literal name | Literal value | Description        |
|--------------|---------------|--------------------|
| SWSelfLock   | 0             | SW Self Lock       |
| VBattEn      | 1             | Enable Battery Bus |
| BCROutEn     | 2             | Enable BCR Bus     |

Continued on the following page



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| Literal name | Literal value | Description                                   |
|--------------|---------------|-----------------------------------------------|
| SHD_3v3      | 3             | Enable 3V3 Bus                                |
| SHD_5v       | 4             | Enable 5V Bus                                 |
| LUP3v3       | 5             | Enable 3V3 LUP, High Level OFF                |
| LUP5v        | 6             | Enable 5V LUP, High Level OFF                 |
| SHDChrg      | 7             | Shutdown Battery Charger, High level = OFF    |
| Chrg_I1      | 8             | Fast battery charge, max current 230 mA       |
| Chrg_I2      | 9             | Fast Battery charge, max current 350 mA       |
| OUT1         | 10            | Output 1                                      |
| OUT2         | 11            | Output 2                                      |
| OUT3         | 12            | Output 3                                      |
| OUT4_OBC     | 13            | Output 4, OBC enable                          |
| OUT5         | 14            | Output 5                                      |
| OUT6         | 15            | Output 6                                      |
| Heater_1     | 16            | Battery heater 1                              |
| Heater_2     | 17            | Battery heater 2                              |
| Heater_3     | 18            | Battery heater 3                              |
| I2C_10k_pu   | 19            | Pull-Up resistor for i2c comm, 10 kOhms       |
| I2C_4k7_pu   | 20            | Pull-Up resistor for i2c comm, 4k7 Ohms       |
| RS485_120R   | 21            | Termination resistor for RS485 comm, 120 Ohms |
| Reserved     | 22            | Reserved                                      |
| OUT7         | 23            | Output 7                                      |
| OUT8         | 24            | Output 8                                      |
| RESET_MPPT   | 25            | Reserved for MPPT                             |





## QueryDefaultsIndex

Helper enumeration for accessing "queryDefaults" elements by named index

### Literals

| Literal name | Literal value | Description                                   |
|--------------|---------------|-----------------------------------------------|
| LUP3v3       | 0             | Enable 3V3 LUP, High Level OFF                |
| LUP5v        | 1             | Enable 5V LUP, High Level OFF                 |
| Reserved1    | 2             | Reserved                                      |
| Chrg_I1      | 3             | Fast battery charge, max current 230 mA       |
| Chrg_I2      | 4             | Fast Battery charge, max current 350 mA       |
| Reserved2    | 5             | Reserved                                      |
| Out_1        | 6             | Output 1                                      |
| Out_2        | 7             | Output 2                                      |
| Out_3        | 8             | Output 3                                      |
| Out_4_OBC    | 9             | Output 4                                      |
| Out_5        | 10            | Output 5                                      |
| Out_6        | 11            | Output 6                                      |
| I2C_10k_pu   | 12            | Pull-Up resistor for i2c comm, 10 kOhms       |
| I2C_4k7_pu   | 13            | Pull-Up resistor for i2c comm, 4k7 Ohms       |
| RS485_120R   | 14            | Termination resistor for RS485 comm, 120 Ohms |
| Out_7        | 15            | Output 7                                      |
| Out_8        | 16            | Output 8                                      |
| MAC_Driver   | 17            | Mac Driver Status                             |



## BLoad\_RetCode

Bootloader return codes

### Literals

| Literal name   | Literal value | Description           |
|----------------|---------------|-----------------------|
| Success        | 0             | Success               |
| Not_Erased     | 1             | Not erased            |
| Not_Supported  | 2             | Command not supported |
| Wrong_Addr     | 3             | Wrong Address         |
| Verify_Failure | 4             | Verification Failure  |
| No_Data        | 5             | No data               |
| Decode_Failure | 6             | Decoding failure      |

## WatchDogType

Watchdog type

### Literals

| Literal name | Literal value | Description                                                                                                                                              |
|--------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ignored      | 0             | Used in request only, no changed to mode will be introduced                                                                                              |
| Disabled     | 1             | Watchdog timer never runs                                                                                                                                |
| Payload_Auto | 2             | Watchdog timer will run after first command 'Feed the dog' ("resetWatchdog"), on Expired event the payload will be turned OFF for a short period of time |
| Payload_On   | 3             | Watchdog timer will run after system startup is finished, on Expired event the payload will be turned OFF for a short period of time                     |

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| Literal name | Literal value | Description                                                                                                                                                |
|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System_Auto  | 4             | Watchdog timer will runs after first command 'Feed the dog' ("resetWatchdog"), on Expired event the whole system restarts, power-on counter is incremented |
| System_On    | 5             | Watchdog timer will runs after system startup is finished, on Expired event the whole system restarts, power-on counter is incremented                     |

## Reset\_RetCode

Reset return codes

## Literals

| Literal name   | Literal value | Description                                 |
|----------------|---------------|---------------------------------------------|
| Success        | 0             | Request is pending                          |
| Wrong_Security | 1             | Wrong security key                          |
| Busy           | 2             | Other reset request is currently processing |

## BatteryInfo

The EPS1 Response structure for Battery Info Request

## Attributes

| Attribute name | Attribute type | Description                                  |
|----------------|----------------|----------------------------------------------|
| batt           | VA_Pair        | Batt. BUS voltage/current - 12 bit ADC Units |
| bcr            | VA_Pair        | Ext. Battery Pack - 12 bit ADC Units         |

Continued on the following page



Continued from the previous page

| Attribute name | Attribute type | Description                                            |
|----------------|----------------|--------------------------------------------------------|
| reserved       | UInt16         | For internal use only. Calibration:<br>*2.3394775 [mV] |

### BusesInfo

The EPS1 Response structure for Buses Info Request

#### Attributes

| Attribute name | Attribute type | Description                                       |
|----------------|----------------|---------------------------------------------------|
| bus_3v3        | VA_Pair        | 3V3 BUS voltage/current - 12 bit ADC Units        |
| bus_5v         | VA_Pair        | 5V BUS voltage/current - 12 bit ADC Units         |
| lup_3v3_volt   | UInt16         | 3V3 Latch-up Protected voltage - 12 bit ADC Units |
| lup_5v_volt    | UInt16         | 5V Latch-up Protected voltage - 12 bit ADC Units  |

### SolarPanelsInfo

The EPS1 Response structure for Solar Panels Info Request

#### Attributes

| Attribute name | Attribute type | Description                                           |
|----------------|----------------|-------------------------------------------------------|
| x              | VAA_Pack       | Solar Panel X axis voltage/current - 12 bit ADC Units |
| y              | VAA_Pack       | Solar Panel Y axis voltage/current - 12 bit ADC Units |
| z              | VAA_Pack       | Solar Panel Z axis voltage/current - 12 bit ADC Units |



## StatisticInfo

The EPS1 Response structure for Statistic Info Request

### Attributes

| Attribute name | Attribute type          | Description                                                                    |
|----------------|-------------------------|--------------------------------------------------------------------------------|
| counters       | Statistic_-<br>Counters | Power on, charge, undervoltage, shortcut circuit and over temperature counters |
| maxTemp        | UInt16[4]               | Batt. Pack temperature sensors with calibration identically to MCU temperature |
| minTemp        | UInt16[4]               | Batt. Pack temperature sensors with calibration identically to MCU temperature |
| rbat_initial   | UInt16                  | Calibration: *1.4972656 [mOhm]                                                 |
| rbat_current   | UInt16                  | Calibration: *1.4972656 [mOhm]                                                 |
| reserved       | UInt16                  | For future use of external temp. sensor                                        |
| uptime         | UInt32                  | Time elapsed since last boot.                                                  |

## TemperatureInfo

The EPS1 Response structure for Temperatures Info Request. The returned values correspond to raw values and shall be calibrated by the user through the calibration factors and formulas hereby presented.

### Attributes

| Attribute name | Attribute type | Description                                                                                                                                                                                                                                          |
|----------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mcu            | UInt16         | $((RV \times 2.5/4096) - 0.986) / 0.00355$ [C]                                                                                                                                                                                                       |
| battery        | UInt16[8]      | up to 8 temperatures read out 1..4 are battery temperature sensors, 5..7 are external temperature sensors, 8 reserved.<br>Positive value calibration: *0.00390625,<br>Negative value calibration: $((RV \gg 4) - 1) \text{XOR } 0xFFFF(-0.0624)$ [C] |



## ConfigurationInfo

The EPS1 Response structure for Configuration Info Request

### Attributes

| Attribute name   | Attribute type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inputConditions  | UInt16         | Bitfield - Input conditions Bit 0 corresponds to Kill switches Lock and Statistics Reset Bits 1 to 15 are note in use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| outputConditions | UInt16[2]      | Bitfield - Output conditions, 2x uint16, outputConditions[0] Bit 0 - SW Self Lock Bit 1 - Enable Battery BUS Bit 2 - Enable BCR BUS Bit 3 - Enable 3.3V BUS Bit 4 - Enable 5V BUS Bit 5 - RESERVED Bit 6 - RESERVED Bit 7 - Output 1 Bit 8 - Output 2 Bit 9 - Output 3 Bit 10 - Output 4/OBC Bit 11 - Output 5/UHF Bit 12 - Output 6 Bit 13 - Battery heater 1 Bit 14 - Battery heater 2 Bit 15 - Battery heater 3 outputConditions[1] Bit 0 - Latch-up Protected output 3.3V (High Level - OFF) Bit 1 - Latch-up Protected output 5V (High Level - OFF) Bit 2 - Shutdown Battery Charger (High Level - OFF) Maximum Current 230mA. Bit 3 - Fast battery charge. Maximum current limit is increased with 230mA Bit 4 - Fast battery charge. Maximum current limit is increased with 230mA Bit 5 - Pull up resistor for I2C communication (10k ohms) Bit 6 - Pull up resistor for I2C communication (4.7k ohms) Bit 7 - Terminating resistor for RS485 communication (120 ohms) Bit 8 - Output 7 Bit 9 - Output 8 Bit 10 - RESERVED Bit 11 - RESERVED Bit 12 - RESERVED Bit 13 - RESERVED Bit 14 - RESERVED Bit 15 - RESERVED |

Continued on the following page



Continued from the previous page

| Attribute name | Attribute type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| defaultOutputs | UInt16[2]      | Bitfield - Default Output states, 2x uint16, defaultOutputs[0] Bit 0 - LUP 3.3V Default Bit 1 - LUP 5V Default Bit 2 - Fast Charge +230mA Default Bit 3 - Fast Charge +230mA Default Bit 4 - I2C_PULLUP_RES_10K Bit 5 - 2C_PULLUP_RES_4K7 Bit 6 - RS485_120ohm_TERMINATING_RES Bit 7 - RESERVED Bit 8 - RESERVED Bit 9 - RESERVED Bit 10 - RESERVED Bit 11 - RESERVED Bit 12 - RESERVED Bit 13 - RESERVED Bit 14 - RESERVED Bit 15 - RESERVED defaultOutputs[1] Bit 0 - Output 1 Default Bit 1 - Output 2 Default Bit 2 - Output 3 Default Bit 3 - Output 4 Default Bit 4 - Output 5 Default Bit 5 - Output 6 Default Bit 6 - Output 7 Default Bit 7 - Output 8 Default Bit 8 - RESERVED Bit 9 - RESERVED Bit 10 - RESERVED Bit 11 - RESERVED Bit 12 - RESERVED Bit 13 - RESERVED Bit 14 - RESERVED Bit 15 - RESERVED |

## 6.3 Interface Methods

### 6.3.1 getSoftwareVersion (id: 0x00)

```
method getSoftwareVersion(  
    out  
    {  
        VersionVariant ver_variant,  
        UInt16 ver_number,  
        String buildDate,  
        String buildTime  
    }  
)
```

Acquire the software version information for the EPS 1 module.



## Output arguments

| Argument name | Argument type  | Description                                                                  |
|---------------|----------------|------------------------------------------------------------------------------|
| ver_variant   | VersionVariant | software state                                                               |
| ver_number    | UInt16         | FW version number                                                            |
| buildDate     | String         | timestamp in format: "MMM DD YYYY"("Dec 20 2021" => 20th December 2021 year) |
| buildTime     | String         | timestamp in format: "hh:mm:ss" - hour:minutes:seconds                       |

### 6.3.2 getBatteryInfo (id: 0x01)

```
method getBatteryInfo(  
    out  
    {  
        BatteryInfo battery  
    }  
)
```

Reads the battery information and telemetry of the EPS 1 Module.

## Output arguments

| Argument name | Argument type | Description                                  |
|---------------|---------------|----------------------------------------------|
| battery       | BatteryInfo   | Battery and BCR voltage + current parameters |

### 6.3.3 getBusesInfo (id: 0x02)

```
method getBusesInfo(  
    out  
    {  
        BusesInfo buses  
    }  
)
```

Reads the buses information and telemetry of the EPS 1 module.





### Output arguments

| Argument name | Argument type | Description                                           |
|---------------|---------------|-------------------------------------------------------|
| buses         | BusesInfo     | 3V3, 5V, LUP 3V3, LUP 5V voltage + current parameters |

#### 6.3.4 getSolarPanelsInfo (id: 0x03)

```
method getSolarPanelsInfo(  
    out  
    {  
        SolarPanelsInfo solar  
    }  
)
```

Provides solar panel information and telemetry (voltage/current) parameters as received by the EPS 1 module.

### Output arguments

| Argument name | Argument type   | Description                        |
|---------------|-----------------|------------------------------------|
| solar         | SolarPanelsInfo | Solar panels voltages and currents |

#### 6.3.5 getStatisticInfo (id: 0x04)

```
method getStatisticInfo(  
    out  
    {  
        StatisticInfo statistics  
    }  
)
```

Reads statistic data



## Output arguments

| Argument name | Argument type | Description                          |
|---------------|---------------|--------------------------------------|
| statistics    | StatisticInfo | General statistics and counters info |

### 6.3.6 getTemperaturesInfo (id: 0x05)

```
method getTemperaturesInfo(  
    out  
    {  
        TemperatureInfo temperatures  
    }  
)
```

Reads the temperatures of MCU and batteries

## Output arguments

| Argument name | Argument type    | Description                       |
|---------------|------------------|-----------------------------------|
| temperatures  | Temperature-Info | MCU + Batt temperature parameters |

### 6.3.7 getConfigurationInfo (id: 0x06)

```
method getConfigurationInfo(  
    out  
    {  
        ConfigurationInfo config  
    }  
)
```

Reads the current state of inputs and outputs, and its default values



## Output arguments

| Argument name | Argument type     | Description                                         |
|---------------|-------------------|-----------------------------------------------------|
| config        | ConfigurationInfo | Input/Output/Default conditions of the EPS I module |

### 6.3.8 queryControls (id: 0x07)

```
method queryControls(  
    in  
    {  
        QueryState[26] requested  
    }  
    out  
    {  
        QueryState[26] reported  
    }  
)
```

Changes the outputs state and receives its current states

## Input arguments



| Argument name | Argument type  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requested     | QueryState[26] | Enumeration represents requested state for particular output, See enumerator OutputState. If currently output state mismatch the request, bit "Mismatch" will be set. [ 0] - SWSelfLock [ 1] - VBattEn [ 2] - BCROutEn [ 3] - SHD_3.3V [ 4] - SHD_5V [ 5] - LUP3.3V [ 6] - LUP5V [ 7] - SHDChrg [ 8] - Chrg_I1 [ 9] - Chrg_I2 [10] - OUT1 [11] - OUT2 [12] - OUT3 [13] - OUT4/OBC [14] - OUT5 [15] - OUT6 [16] - Heater 1 [17] - Heater 2 [18] - Heater 3 [19] - 10k I2C p/u [20] - 4k7 I2C p/u [21] - RS485 120R [22] - reserved1 [23] - OUT7 [24] - OUT8 [25] - RESET_MPPT |

### Output arguments

| Argument name | Argument type  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reported      | QueryState[26] | Enumeration represents requested state for particular output, See enumerator OutputState [ 0] - SWSelfLock [ 1] - VBattEn [ 2] - BCROutEn [ 3] - SHD_3.3V [ 4] - SHD_5V [ 5] - LUP3.3V [ 6] - LUP5V [ 7] - SHDChrg [ 8] - Chrg_I1 [ 9] - Chrg_I2 [10] - OUT1 [11] - OUT2 [12] - OUT3 [13] - OUT4/OBC [14] - OUT5 [15] - OUT6 [16] - Heater 1 [17] - Heater 2 [18] - Heater 3 [19] - 10k I2C p/u [20] - 4k7 I2C p/u [21] - RS485 120R [22] - reserved1 [23] - OUT7 [24] - OUT8 [25] - RESET_MPPT |

### 6.3.9 queryRBatRaw (id: 0x08)

```
method queryRBatRaw(  
    in  
    {  
        UInt8 newValue  
    }  
    out
```



```
{
    UInt8 currentValue
}
```

Sets the initial value of the resistance of the battery

#### Input arguments

| Argument name | Argument type | Description                                                                                                                   |
|---------------|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| newValue      | UInt8         | new value of battery pack resistance. Values 0 and >0xFF will do no changes (but the commands will return the current value). |

#### Output arguments

| Argument name | Argument type | Description                         |
|---------------|---------------|-------------------------------------|
| currentValue  | UInt8         | Current value of batter resistance. |

#### 6.3.10 queryDefaults (id: 0x09)

```
method queryDefaults(
    in
    {
        QueryState[17] requested,
        UInt8 reserved_req
    }
    out
    {
        QueryState[17] reported,
        UInt8 reserved_resp
    }
)
```

Changes the default outputs state and receives its current default states after executing the command



## Input arguments

| Argument name | Argument type  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requested     | QueryState[17] | Enumeration represents requested state for particular output, See enumerator OutputState, valid states are ON and OFF, any value other than ON/OFF will be ignored. any value for Reserved fields will be ignored, however recommendation is to send 0. Some values are effective immediately, some are after reset only! [ 0] - LUP3v3 [ 1] - LUP5v [ 2] - Reserved [ 3] - Fast_charge1_230mA [ 4] - Fast_charge2_230mA [ 5] - Reserved [ 6] - Out_1 [ 7] - Out_2 [ 8] - Out_3 [ 9] - Out_4/OBC [10] - Out_5 [11] - Out_6 [12] - 10k I2C pull up [13] - 4k7 I2C pull up [14] - RS485 120R ohms termination resistor [15] - Out_7 [16] - Out_8 |
| reserved_req  | UInt8          | For internal Use only. Keep the value as 0x00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Output arguments



| Argument name | Argument type  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reported      | QueryState[17] | Enumeration represents requested state for particular output, See enumerator OutputState, valid states are ON/OFF only, "Ignored" for reserved MAC driver: also ON/OFF values will be returned, flag "Mismatch" will indicates non-default driver running now. [ 0] - LUP3v3 [ 1] - LUP5v [ 2] - Reserved [ 3] - Fast_charge1_230mA [ 4] - Fast_charge2_230mA [ 5] - Reserved [ 6] - Out_1 [ 7] - Out_2 [ 8] - Out_3 [ 9] - Out_4/OBC [10] - Out_5 [11] - Out_6 [12] - 10k I2C pull up [13] - 4k7 I2C pull up [14] - RS485 120R ohms termination resistor [15] - Out_7 [16] - Out_8 |
| reserved_resp | UInt8          | For internal Use only. Keep the value as 0x00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### 6.3.11 queryLaunchState (id: 0x0A)

```
method queryLaunchState(  
    in  
    {  
        QueryState requested,  
        UInt8 writeDelayPeriod  
    }  
    out  
    {  
        QueryState reported,  
        UInt8 readDelayPeriod  
    }  
)
```

Changes the default outputs state and receives its current states



### Input arguments

| Argument name    | Argument type | Description                                                                                                                                                                                                                                                      |
|------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requested        | QueryState    | Enumeration represents requested state for next power-on cycle - would it be launch (by default 5 minutes delay before all power buses will be enabled), or not (immediately) Valid Input values are "On" and "Off", all values other than that will be ignored. |
| writeDelayPeriod | UInt8         | Number of minutes to wait until providing power supply after lunch. That works only if the function is enabled by the above parameter.                                                                                                                           |

### Output arguments

| Argument name   | Argument type | Description                                                          |
|-----------------|---------------|----------------------------------------------------------------------|
| reported        | QueryState    | reported State with the new default value                            |
| readDelayPeriod | UInt8         | number of minutes to wait until providing the new value after reboot |

#### 6.3.12 queryWatchdog (id: 0x0B)

```
method queryWatchdog(  
    in  
    {  
        UInt8 new_timer,  
        WatchDogType new_type  
    }  
    out  
    {  
        UInt8 remaining_time,  
        WatchDogType type  
    }  
)
```

Get/Set watchdog configuration - timer and output mask to reset





### Input arguments

| Argument name | Argument type | Description                                                                                                                                                                                                                                                                           |
|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| new_timer     | UInt8         | Watchdog timer period in seconds, minimum 10 seconds NOTE 1: values below 10 will be ignored, and could be used to obtain current timer value NOTE 2: if Auto mode is set, the watchdog will be stopped after timer value is set, even if it was already running in auto mode before! |
| new_type      | WatchDogType  | See WatchDogType; BE AWARE: if Auto mode is set, the watchdog will not run automatically, even if it was already running in auto mode before!                                                                                                                                         |

### Output arguments

| Argument name  | Argument type | Description                                       |
|----------------|---------------|---------------------------------------------------|
| remaining_time | UInt8         | returns currently set watchdog timer and its type |
| type           | WatchDogType  | returns new set watch dog type                    |

#### 6.3.13 resetWatchdog (id: 0x0C)

```
method resetWatchdog(  
    out  
    {  
        WatchDogType type,  
        UInt8 timer  
    }  
)
```

Resets the Watchdog timer in order to prevent MCU Reset on its timeout



## Output arguments

| Argument name | Argument type | Description                                                                                                                                                                                       |
|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type          | WatchDogType  | 'type' will let you know about current watchdog settings                                                                                                                                          |
| timer         | UInt8         | 'timer' is a time left until reset before the reset action, in other words: how much time you have left to watchdog expired. if no watchdog is running, the value of 'watchdog timer' is returned |

### 6.3.14 resetPowerSupply (id: 0x0D)

```
method resetPowerSupply(  
    in  
    {  
        UInt8 security  
    }  
    out  
    {  
        Reset_RetCode result  
    }  
)
```

Resets the MCU and/or all the outputs the delay of 100ms before reset is implemented in order to confirm the command. however, on a high ESPS load it is possible that response will not be sent due to time limits.

## Input arguments



| Argument name | Argument type | Description                                                                                                                                                                                                                                                                                                                                                        |
|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| security      | UInt8         | Security key to avoid accidental resets.<br>security = 0x00 - Payload reset (power failure simulation), security = 0x12 - Soft MCU reset - wait for critical tasks to be done first, security = 0x5A - Hard MCU reset - reset the MCU faster, however still try to send response, security = 0xDE - 'Kill It!' - reset MCU ASAP, no response is almost guaranteed. |

### Output arguments

| Argument name | Argument type | Description        |
|---------------|---------------|--------------------|
| result        | Reset_RetCode | reset code returns |

#### 6.3.15 SetSerialNumber (id: 0x0E)

```
method SetSerialNumber(  
    in  
    {  
        String DeviceSerialNumberWrite  
    }  
    out  
    {  
        String DeviceSerialNumberRead  
    }  
)
```

Write (once per lifetime) and Read afterwards the EPS1 serial number



### Input arguments

| Argument name           | Argument type | Description            |
|-------------------------|---------------|------------------------|
| DeviceSerialNumberWrite | String        | Serial number to write |

### Output arguments

| Argument name          | Argument type | Description        |
|------------------------|---------------|--------------------|
| DeviceSerialNumberRead | String        | EPS1 Serial number |

#### 6.3.16 GetSerialNumber (id: 0x0F)

```
method GetSerialNumber(  
    out  
    {  
        String DeviceSerialNumber  
    }  
)
```

Read the EPS1 serial number

### Output arguments

| Argument name      | Argument type | Description        |
|--------------------|---------------|--------------------|
| DeviceSerialNumber | String        | EPS1 Serial number |

#### 6.3.17 HIBModeRead (id: 0x10)

```
method HIBModeRead(  
    out  
    {  
        UInt8 entryLevelVoltage ,  
        UInt8 exitLevelVoltage ,  
        UInt8 entryLevelTemperature ,  
        UInt8 exitLevelTemperature
```



```
}  
)
```

Read the Settings of HIB mode

### Output arguments

| Argument name         | Argument type | Description                                                                                                                                           |
|-----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| entryLevelVoltage     | UInt8         | Battery voltage threshold under which will enter HIB mode. The value should be divided by calibration constant 53.43 (example: $187 / 53.43 = 3.5V$ ) |
| exitLevelVoltage      | UInt8         | Battery voltage threshold under which will exit HIB mode. The value should be divided by calibration constant 53.43 (example: $187 / 53.43 = 3.5V$ )  |
| entryLevelTemperature | UInt8         | Temperature over which will enter HIB mode. The value is integer in degree Celsius                                                                    |
| exitLevelTemperature  | UInt8         | Temperature under which will enter HIB mode. The value is integer in degree Celsius                                                                   |

### 6.3.18 HIBModeWrite (id: 0x11)

```
method HIBModeWrite(  
    in  
    {  
        UInt8 writeEntryLevelVoltage ,  
        UInt8 writeExitLevelVoltage ,  
        UInt8 writeEntryLevelTemperature ,  
        UInt8 writeExitLevelTemperature  
    }  
    out  
    {  
        UInt8 readEntryLevelVoltage ,  
        UInt8 readExitLevelVoltage ,  
        UInt8 readEntryLevelTemperature ,
```



```
        UInt8 readExitLevelTemperature
    }
)
```

Write the Settings of HIB mode

### Input arguments

| Argument name              | Argument type | Description                                                                                                                                                     |
|----------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| writeEntryLevelVoltage     | UInt8         | Battery voltage threshold under which will enter HIB mode. The value should be multiplied by calibration constant 53.43 (example: $3.5V \times 53.43 = 187$ )   |
| writeExitLevelVoltage      | UInt8         | Battery voltage threshold under which will exit HIB mode. The value should be multiplied by calibration constant 53.43 (example: $3.5V \times 53.43 = 187$ )    |
| writeEntryLevelTemperature | UInt8         | Temperature over which will enter HIB mode. The value is integer in degree Celsius (example: to write 2.2 degC -> round down (or up) and set value = 2 (or 3))  |
| writeExitLevelTemperature  | UInt8         | Temperature under which will enter HIB mode. The value is integer in degree Celsius (example: to write 2.2 degC -> round down (or up) and set value = 2 (or 3)) |

### Output arguments

| Argument name         | Argument type | Description                                                                                                                                           |
|-----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| readEntryLevelVoltage | UInt8         | Battery voltage threshold under which will enter HIB mode. The value should be divided by calibration constant 53.43 (example: $187 / 53.43 = 3.5V$ ) |

Continued on the following page



Continued from the previous page

| Argument name             | Argument type | Description                                                                                                                                          |
|---------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| readExitLevelVoltage      | UInt8         | Battery voltage threshold under which will exit HIB mode. The value should be divided by calibration constant 53.43 (example: $187 / 53.43 = 3.5V$ ) |
| readEntryLevelTemperature | UInt8         | Temperature over which will enter HIB mode. The value is integer in degree Celsius                                                                   |
| readExitLevelTemperature  | UInt8         | Temperature under which will enter HIB mode. The value is integer in degree Celsius                                                                  |

### 6.3.19 HeatersSettingsRead (id: 0x12)

```
method HeatersSettingsRead(  
    out  
    {  
        Int8[3] lowBatTemperatureHeater,  
        UInt8[3] hysteresisHeater  
    }  
)
```

Read the Settings of all three heaters

#### Output arguments

| Argument name           | Argument type | Description                                                            |
|-------------------------|---------------|------------------------------------------------------------------------|
| lowBatTemperatureHeater | Int8[3]       | Battery temperature under which will turn On each of the three heaters |
| hysteresisHeater        | UInt8[3]      | Hysteresis to turn Off the heaters                                     |



### 6.3.20 HeatersSettingsWrite (id: 0x13)

```
method HeatersSettingsWrite(  
    in  
    {  
        Int8[3] writeLowBatTemperatureHeater ,  
        UInt8[3] writeHysteresisHeater  
    }  
    out  
    {  
        Int8[3] readLowBatTemperatureHeater ,  
        UInt8[3] readHysteresisHeater  
    }  
)
```

Write and Read afterwards the Settings of all heaters

#### Input arguments

| Argument name                 | Argument type | Description                                              |
|-------------------------------|---------------|----------------------------------------------------------|
| writeLowBatTemperature-Heater | Int8[3]       | Battery temperature under which will turn On the heaters |
| writeHysteresisHeater         | UInt8[3]      | Hysteresis to turn Off the heaters                       |

#### Output arguments

| Argument name                | Argument type | Description                                              |
|------------------------------|---------------|----------------------------------------------------------|
| readLowBatTemperature-Heater | Int8[3]       | Battery temperature under which will turn On the heaters |
| readHysteresisHeater         | UInt8[3]      | Hysteresis to turn Off the heaters                       |





### 6.3.21 ToggleOutPut (id: 0x14)

```
method ToggleOutPut(  
    in  
    {  
        QueryControlsIndex toggleOutput,  
        UInt8 timeAtToggledState  
    }  
    out  
    {  
        Boolean requestStatus,  
        UInt8 actualTimeAtToggledState  
    }  
)
```

Changed the state of one output and after the toggle timeout will set it back to its original state

#### Input arguments

| Argument name      | Argument type           | Description                                                                                                                   |
|--------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| toggleOutput       | QueryCon-<br>trolsIndex | Output number to be toggled                                                                                                   |
| timeAtToggledState | UInt8                   | Time in seconds for the output to stay toggled state. If 0 - that will be ignored and last settings (or default) will be used |

#### Output arguments

| Argument name            | Argument type | Description                             |
|--------------------------|---------------|-----------------------------------------|
| requestStatus            | Boolean       | status of the request, 1- done, 0- fail |
| actualTimeAtToggledState | UInt8         | time to stay at the toggled state       |



### 6.3.22 NotificationsSettings (id: 0x15)

```
method NotificationsSettings(  
    in  
    {  
        UInt8 repetitionsPeriod,  
        UInt8 numberRepetitions  
    }  
    out  
    {  
        UInt8 actualRepetitionsPeriod,  
        UInt8 actualNumberRepetitions  
    }  
)
```

The EPS I will send broadcast commands to notify other devices for an event that may result in power down of all payloads.

#### Input arguments

| Argument name     | Argument type | Description                                                                   |
|-------------------|---------------|-------------------------------------------------------------------------------|
| repetitionsPeriod | UInt8         | repetition time - time between two notifying broadcasts; To read send value 0 |
| numberRepetitions | UInt8         | number of repetitions for one notification; To read send value 0              |

#### Output arguments

| Argument name           | Argument type | Description                                             |
|-------------------------|---------------|---------------------------------------------------------|
| actualRepetitionsPeriod | UInt8         | repetition time - time between two notifying broadcasts |
| actualNumberRepetitions | UInt8         | number of repetitions for one notification              |



### 6.3.23 GetTelemetryAltogether (id: 0x16)

```
method GetTelemetryAltogether(  
    out  
    {  
        BatteryInfo battery,  
        BusesInfo buses,  
        SolarPanelsInfo solar,  
        TemperatureInfo temperatures,  
        ConfigurationInfo config,  
        UInt8 RBatRaw_currentValue  
    }  
)
```

some of the previous commands combined

#### Output arguments

| Argument name        | Argument type          | Description                                                |
|----------------------|------------------------|------------------------------------------------------------|
| battery              | BatteryInfo            | The EPS1 Response structure for Battery Info Request       |
| buses                | BusesInfo              | The EPS1 Response structure for Buses Info Request         |
| solar                | SolarPanelsInfo        | The EPS1 Response structure for Solar Panels Info Request  |
| temperatures         | Temperature-Info       | The EPS1 Response structure for Temperatures Info Request. |
| config               | Configura-<br>tionInfo | The EPS1 Response structure for Configuration Info Request |
| RBatRaw_currentValue | UInt8                  | Battery current raw value                                  |



#### 6.3.24 `bload_getVersion` (id: 0x10000000)

```
method bload_getVersion(  
    out  
    {  
        UInt8 revision,  
        UInt8 supported  
    }  
)
```

Gets the current Bootloader version, if there is any

##### Output arguments

| Argument name | Argument type | Description                                                                                                                                                 |
|---------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| revision      | UInt8         | Bootloader revision (0 for A, 1 for B, etc.)                                                                                                                |
| supported     | UInt8         | Bitmask of supported image types bit 0 corresponds to 'Plain Text', bit 1 corresponds to 'EnduroSat Crypto v1', other bits are reserved for next revisions. |

#### 6.3.25 `bload_eraseFlash` (id: 0x10000001)

```
method bload_eraseFlash(  
    in  
    {  
        UInt32 reqCode  
    }  
    out  
    {  
        BLoad_RetCode retCode  
    }  
)
```

Command to erase the storage flash memory for writing the new image of the application



### Input arguments

| Argument name | Argument type | Description                                                                                                                                                                                         |
|---------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reqCode       | UInt32        | 0x00000000 - Check if flash is empty<br>0x12345678u - erase the flash memory and set image type to "Plain Text"<br>0x74615345u - erase the flash memory and set image type to "EnduroSat Crypto v1" |

### Output arguments

| Argument name | Argument type  | Description             |
|---------------|----------------|-------------------------|
| retCode       | BLoad_-RetCode | Bootloader return codes |

#### 6.3.26 `bload_writeFlash` (id: 0x10000002)

```
method bload_writeFlash(  
    in  
    {  
        UInt16 address,  
        UInt8[128] data  
    }  
    out  
    {  
        BLoad_RetCode retCode  
    }  
)
```

Writes part of a image file of a new application in the storage flash memory



### Input arguments

| Argument name | Argument type | Description                                                  |
|---------------|---------------|--------------------------------------------------------------|
| address       | UInt16        | address in memory to write the specific block. Must be even. |
| data          | UInt8[128]    | Data memory to write, up to 128 bytes. Must be even.         |

### Output arguments

| Argument name | Argument type     | Description             |
|---------------|-------------------|-------------------------|
| retCode       | BLoad_<br>RetCode | Bootloader return codes |

#### 6.3.27 `bload_execute` (id: 0x10000003)

```
method bload_execute(  
    in  
    {  
        UInt32 CRC32  
    }  
    out  
    {  
        BLoad_RetCode retCode  
    }  
)
```

Request the installation of the new image from the storage flash over the current application



### Input arguments

| Argument name | Argument type | Description                                                                                 |
|---------------|---------------|---------------------------------------------------------------------------------------------|
| CRC32         | UInt32        | Cyclic redundancy check of the image of the new firmware. Poly: 0x04C11DB7L. Reverse : TRUE |

### Output arguments

| Argument name | Argument type     | Description             |
|---------------|-------------------|-------------------------|
| retCode       | BLoad_<br>RetCode | Bootloader return codes |

## 6.4 EPS I Notify Interface Methods Overview

| Method name        | Method description                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EventsNotification | Notify on event of the EPS by sending broadcast commands depending on the "NotificationsSettings" from "EPS1" interface. The EPS will try to postpone power down until all notifying repetitions are sent. In order to protect the EPS, if a timeout is reached actions will be taken even if not all repetitions are sent. The timeouts depend on the type of each event. |

## 6.5 Data Types

### eEventNumber

List of events that the device is monitoring and notifying all devices on the bus about them.



## Literals

| Literal name               | Literal value | Description                                                                            |
|----------------------------|---------------|----------------------------------------------------------------------------------------|
| HIB_MODE_BAT_CRIT_-LVL     | 0             | Critical battery level. Follows switching to "HIB mode"                                |
| HIB_MODE_HIGH_-TEMPERATURE | 1             | Too High Temperature. Follows switching to "HIB mode"                                  |
| LOW_BAT                    | 2             | Notification for low battery level. Entering to critical level may be expected.        |
| WATCHDOG_IMMEDIATE_-RESET  | 3             | Reset of the whole system (including the EPS) that is requested by an external command |
| WATCHDOG_PAYLOAD_-RESET    | 4             | Reset of all payloads that is requested by an external command                         |

## 6.6 Interface Methods

### 6.6.1 EventsNotification (id: 0x01)

```
method EventsNotification(  
    in  
    {  
        eEventNumber Mode,  
        UInt8 timeout  
    }  
)
```

Notify on event of the EPS by sending broadcast commands depending on the "Notification-Settings" from "EPS1" interface. The EPS will try to postpone power down until all notifying repetitions are sent. In order to protect the EPS, if a timeout is reached actions will be taken even if not all repetitions are sent. The timeouts depend on the type of each event.





## Input arguments

| Argument name | Argument type | Description |
|---------------|---------------|-------------|
| Mode          | eEventNumber  |             |
| timeout       | UInt8         |             |

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